What Job Application Data Reveal About Accounting Rookies' Research Potential

Bin Ke[1] and Zhen Long[2]

March 20, 2026

ABSTRACT

Most organizations adopt a structured recruitment funnel that begins with résumé-based screening. Although systematic, this initial screening is time-consuming and error-prone due to limited information and potential human biases. Using the accounting rookie job market as our empirical setting, we examine whether the limited information available at the initial stage can support the development of an effective machine learning algorithm for identifying candidates with stronger research potential. We find that each of the three available inputs—résumés, the quality of prior educational institutions, PhD advisors, and coauthors, and the job market paper—independently predicts subsequent research productivity. When combined, these inputs generate a screening algorithm that performs two to four times better than random guesses. We further document that the algorithm exhibits lower bias relative to human recruiters. Taken together, our findings show the promise of algorithmic screening in improving the efficiency and effectiveness of initial-stage applicant screening.

JEL: C53; M12; M41; M51
Keywords: rookie candidates; resume; research productivity; machine learning

We thank K. Ramesh, Shiva Sivaramakrishnan, Lijun Xia, Sara Toynbee, Qi Zhang, Haifeng You, Yong Yu, workshop participants at the NUS, Rice University, University of Macau, Tsinghua University, UT-Austin, Chinese University of Hong Kong, Shanghai Jiaotong University, Zhejiang University, University of Houston, IDEI AI for Business Conference at HKU, for helpful comments. Bin Ke acknowledges financial support from the National University of Singapore and the National Natural Science Foundation of China (grant No. 72442013). This article is written with the assistance of ChatGPT.

[1] NUS Business School, National University of Singapore, Mochtar Riady Building, BIZ 1, #07-30, 15 Kent Ridge Drive, Singapore 119245. Tel: +65 66013133. Fax: +65 67736493. Email: bizk@nus.edu.sg.
[2] School of Accountancy, Zhongnan University of Economics and Law. Email: longzhen@zuel.edu.cn.

1. Introduction

Employee selection is a fundamental component of management control systems.[1] Most organizations follow a structured recruitment funnel, which typically includes multiple stages: initial screening by human recruiters based on limited information, typically only standardized résumés (without direct interaction), followed by several rounds of interviews to collect additional information to evaluate the shortlisted candidates' hard and soft skills. While systematic, the standard recruitment funnel is often time-consuming, costly, and prone to suboptimal outcomes (Boushey and Glynn 2012). One significant hurdle is the large volume of applicants, which pressures recruiters to make quick, superficial decisions during the initial screening phase (Li, Raymond, and Bergman 2026; Lerner and Bergman 2024). This rushed evaluation often results in overlooking qualified candidates, an error that subsequent stages of the recruitment process cannot rectify effectively (Lerner and Bergman 2024). This study proposes and validates a novel personnel selection method designed to enhance both the efficiency and effectiveness of applicant screening in the initial stage of the recruitment funnel.

Traditional employee recruitment processes largely depend on the judgment of human recruiters, which introduces two key limitations (Hoffman, Kahn, and Li 2018; Kahneman, Sibony, and Sunstein 2021). First, recruiters are prone to heuristic and cognitive biases, leading to inconsistent evaluations of job applicants' fit and performance potential. Lerner and Bergman (2024) observe that recruiters perform only marginally better than random chance in evaluating resumes during the initial

[1] Merchant and Van der Stede (2017) classify management control systems into action, results, and personnel and cultural controls. Employee selection is a critical part of personnel control.

stage of employee selection, with significant disagreement on what constitutes a strong candidate. Second, agency conflicts, such as personal preferences or nepotism, can distort judgments. These deficiencies frequently result in mismatches between hired employees and optimal candidates, ultimately hampering firm performance.

To address the inherent limitations of human recruiters, we leverage machine learning techniques (Mullainathan 2025). While machine learning's superior information-processing capabilities relative to human judgment are well established (e.g., Ding, Lev, Peng, Sun, and Vasarhelyi 2020; Liu 2022), it remains an open question whether machine learning–based selection methods are effective when constrained by the limited information available at the initial screening stage.

We assess the feasibility of our proposed selection method using data from the accounting rookie faculty job market. The academic hiring context allows us to address a key obstacle in implementing and validating our predictive selection models: the lack of outcome data for non-hired applicants. In typical recruitment settings, post-hire outcomes are observed only for selected candidates, introducing selection bias and limiting the generalizability of model performance—particularly in contexts with low hiring rates. In contrast, we observe subsequent publication outcomes for all job market candidates, regardless of placement. This feature enables us to train a model on the full set of applicants, reducing selection bias and enhancing predictive validity. In addition, our research question is equally relevant to academic institutions, as they face selection challenges analogous to those faced by corporate recruiters.

Our sample comprises complete application records for 569 candidates—fewer than 150 per year—who participated in the Miami rookie market between 2015 and 2018. The sample ends in 2018 to allow for a 5-year window for measuring post-graduation research performance. Although recruiters assess candidates on multiple dimensions, we focus on research potential because it is both critical to faculty hiring decisions and inherently difficult to measure. Data on other dimensions, such as teaching ability, are unavailable at the initial recruitment stage and hence cannot be analyzed in this study. We expect that recruiters evaluate additional dimensions in subsequent stages of the hiring process, when richer information can be obtained through direct interactions with shortlisted candidates.

We proxy for research potential using the number of publications in 11 leading journals spanning accounting, finance, and management science—outlets widely regarded as central to the accounting discipline. We track publications over a five-year window beginning on January 1 of the candidate's graduation year, reflecting the typical timing of the academic job market, which commences in December. To simplify the prediction task, we convert the continuous publication count into a binary outcome, defined using thresholds at the 5th, 10th, 20th, and 30th percentiles for each year.

To evaluate the predictive value of different types of application data, we construct a series of machine learning models for predicting research productivity using three primary data sets. The first set (denoted as set A) consists of information extracted directly from candidates' resumes. The second set (denoted as set B) includes metrics related to the quality of journals in which the candidates have published, as well as the reputations of their educational institutions, dissertation advisors, and coauthors. While the variables in the second set are not directly included in the candidates' application materials, they are readily obtainable from publicly available sources.

Reading job market papers is one of the most time-consuming tasks for faculty recruiters. We therefore examine whether machine learning can extract useful information from job market papers, which form our third dataset (Set C). To represent paper content, we use OpenAI's large language model, text-embedding-3-large. Given the model's token limit, we restrict the embedded text to each paper's title, abstract, and introduction.[2]

To address potential selection bias resulting from the endogenous matches between job candidates and academic institutions, we classify institutions as research-intensive or non-research-intensive and implement inverse probability weighting (IPW), following Swaminathan and Joachims (2015) and Schnabel et al. (2016). IPW reweights observations by the estimated probability of a given candidate–institution type match, thereby mitigating bias arising from non-random matching.

We begin by evaluating the predictive value of each data source—A, B, and C—when used independently in our machine learning models. We use a cumulative rolling window approach, training models on at least two years of data, and accordingly, the testing samples cover the 2017 and 2018 batches. Given the small training sample sizes, we employ the logit model, which outperforms more advanced methods such as Random Forest in our study.

[2] Accounting rookie candidates may sometimes include the title and abstracts of their job market paper and other working papers within their resumes. However, the resume data in Set A explicitly excludes any information related to such abstract information. As a result, there is no overlap between Sets A and C in our analysis.

For both test years, the research productivity model based on each data source significantly outperforms the unconditional mean benchmark for each dependent variable, which represents random guessing. These results suggest that each data source contains useful information for predicting research productivity.

Next, we evaluate the performance of a research productivity model that combines all three data sources into a single specification (Model ABC). As expected, Model ABC significantly outperforms the unconditional mean benchmark for each dependent variable. Moreover, its performance is generally comparable to, and in many cases exceeds, that of models based on a single data source.

An important question is whether our most comprehensive research productivity specification, Model ABC, outperforms human recruiters when research productivity is used as the sole evaluation criterion. Institutions with higher rankings tend to attract stronger candidates. Moreover, rookie candidates selected by top-ranked institutions often benefit from these institutions' superior resources, which can further enhance their productivity. Consequently, if the profession efficiently matches candidate quality to institutional quality, the collective outcomes of human recruiters should, in theory, outperform the predictions of our machine learning model.[3]

To test this prediction, we rank the universities based on U.S. News & World Report's top 50 graduate business schools, since most accounting candidates in our sample graduated from North American institutions. Institutions not on this list are ranked using the QS World University Ranking,

[3] Since we do not have access to the list of applicants for individual institutions, we cannot evaluate the performance of our best machine learning model at the individual institution level.

while those absent from both rankings are placed at the bottom. To reduce the potential measurement error of the university rankings, we aggregate universities included in the U.S. News and QS World University rankings into deciles for each year. Institutions not appearing on either list are placed in a separate category, ranked at the bottom.

To evaluate the performance of human recruiters across the accounting profession, we train a machine learning model (also a logit model) that uses the decile ranking of the initial placement institution as the sole input (denoted as the Human Selection Model). The results indicate that the Human Selection Model outperforms random guessing for approximately 75% of the specifications in the two combined test years. This evidence is consistent with the view that human recruiters possess information or judgment that improves selection outcomes relative to random assignment. However, the performance of Model ABC exceeds that of the Human Selection Model in 75% of the models in the two combined test years. Notably, as the prediction task becomes more demanding—specifically, when applying the 5% or 10% cutoff—Model ABC outperforms the Human Selection Model, with an average improvement of 92%. These results suggest that human recruiters may have substantially misjudged candidates' research potential, or alternatively, that they place weight on non-research productivity factors that justify forgoing the observed large performance gains.

The performance differences between Model ABC and the Human Selection Model suggest that human recruiters weigh the data sources—A, B, and C—differently than our machine learning models do. To better understand these weighting differences, we compare Model ABC, which predicts research productivity, with a new model that uses the same predictors (ABC) but instead predicts a job candidate's initial job placement (the Initial Placement Model). As with Model ABC, we convert the placement rankings into dichotomous variables using the same cutoffs.

We employ Shapley values (SHAP) to quantify the contribution of individual predictors to prediction performance. Given the large number of predictors, we focus on the top ten predictors based on their SHAP values for each model. We find significant differences in both the ranking and the sign of SHAP values between Model ABC and the Initial Placement Model, suggesting that human recruiters weigh the same information sources differently from Model ABC.

Our final analysis examines whether Model ABC exhibits greater biases than human recruiters against specific candidate groups. We compare the predicted probabilities from Model ABC and the predicted probabilities from the Initial Placement Model across two broad categories of candidate attributes: (i) demographic characteristics such as gender and ethnicity (Asian background, European background), and (ii) institution characteristics (PhD program ranking and productivity of dissertation advisors). The results provide evidence consistent with greater bias in recruiters' selection decisions. Specifically, recruiters exhibit a preference for candidates with a European background, those from the top 10 PhD programs, and those advised by highly productive dissertation supervisors.

The remainder of the paper is structured as follows: Section 2 reviews the prior literature and outlines our contributions. Section 3 details our research design and methodology. Section 4 presents empirical results and related sensitivity checks. Section 5 leverages our best predictive model to analyze the extent of bias reflected in hiring outcomes. Finally, Section 6 concludes with a summary of the findings and implications.

2. Prior literature and contributions

This study contributes to multiple streams of literature across different academic fields. First, we contribute to the literature on managerial accounting and human resource management by addressing a central challenge in employee selection: assessing a candidate’s performance potential based on limited information available at the initial stage of the recruitment process. While most existing studies in accounting and management focus on causal inference questions—such as how to motivate employees through contract design—there is relatively little research on employee selection methods, which are inherently prediction problems.[4] Although prior research has acknowledged the importance of measuring employee performance potential in human capital management (e.g., Deller 2023; Grabner, Künneke, and Moers 2025), few have directly developed predictive methods for measuring human capital performance potential. To the best of our knowledge, Chen and Ke (2026) are among the few studies in the accounting and management literature that employ machine-learning models to assess human capital potential. Chen and Ke (2026) demonstrate the predictive value of resumes in initial applicant screening, but limit their focus to entry-level positions in a financial institution, which require relatively low levels of employee discretion. This raises an important question: can resumes alone effectively predict job performance for positions requiring significant discretion?

[4] Managerial accounting research (Manthei, Sliwka, and Vogelsang 2023) emphasizes two fundamental roles of management control systems—decision-facilitating and decision-influencing—which closely correspond to the concepts of prediction and causal inference, respectively.

We address this question by examining rookie faculty candidates in accounting, whose primary responsibility is to conduct high-quality research—a task requiring substantial creativity and discretion. Using data on candidates' top post-graduation journal publications, we provide evidence that the limited resume information available during initial applicant screening is highly valuable for predicting future research productivity. Moreover, our machine learning models, trained on resume data, significantly outperform human recruiters' selection outcomes.

By integrating the findings from Chen and Ke (2026) and our study, which analyze two distinct organizational contexts and job types, our research underscores the predictive power of resumes in forecasting future job performance. However, our findings also highlight that such information is underutilized in current human-centered recruitment processes. We provide a counterfactual analysis indicating that, if the top 10 institutions were to adopt our most comprehensive model in applicant screening, the concentration of the most productive researchers among these institutions would increase by more than 40%.

Our study responds directly to Kaplan (1998), who advocates for more practice-relevant research through the development of new management tools within organizations, and to Labro (2025), who highlights the scarcity of managerial accounting research offering practical relevance and actionable insights, while calling for greater attention to predictive analytics (see also Casas-Arce, Cheng, Grabner, and Modell 2022). Our study is important because advanced people analytics tools of the type we develop remain underexplored and largely unimplemented in recruitment.

Second, our study contributes to the economics literature on talent selection within academic settings. This literature primarily focuses on causal inference, identifying factors that explain variation in academic researchers' research productivity. Key determinants of research productivity highlighted in prior studies include research funding (Gulbrandsen and Smeby 2005), the rank of the candidate's Ph.D. program (Conley and Önder 2014), advisor's gender (Hussey, Murray, and Stock 2022), parenthood (Krapf, Ursprung, and Zimmermann 2017), coauthorship network (Ductor 2015; Defazio, Lockett, and Wright 2009; Yadav, McHale, and O'Neill 2023; Hilmer and Hilmer 2012; Ductor and Visser 2022), and initial job placement (Oyer 2006).

As expected, most studies in this literature focus on causal inference questions, typically examining associations between academic productivity and one or two specific factors. However, relatively few studies have explored out-of-sample prediction of academic productivity. For instance, Hilmer and Hilmer (2012) collected data on over 3,000 rookie job candidates listed in the Journal of Economic Literature who earned their economics Ph.D. between 1990 and 1994. Using 1994 as a test year, they predicted publication counts based on a limited set of publicly available predictors, given no direct access to candidates' resumes, including six Ph.D. program-tier dummies, advisor prominence, whether the candidate worked with a prominent advisor, and whether they published during graduate school. Similarly, Ductor et al. (2014) constructed a coauthorship network using articles published in EconLit from 1970 to 1999 to investigate how network-related variables predict academic output. Ductor et al. (2014) did not differentiate between rookie and non-rookie scholars, nor did it consider the role of limited resume information at initial applicant screening.

Our study contributes to this emerging literature on prediction by developing a machine learning model tailored to the initial screening of rookie job candidates in accounting—a field closely related to, but distinct from, economics. Importantly, we are the first to evaluate the predictive value of limited job application materials in assessing future research productivity. In addition, our list of predictors is more comprehensive than those used in prior studies, as we leverage the full suite of application information available to human recruiters. This allows us to provide a richer and more realistic assessment of the factors influencing research productivity, advancing the predictive capabilities in academic talent selection.

Third, our study contributes to the growing field of advanced people analytics on person-job fit, a domain primarily driven by researchers in computer engineering.[5] Much of the existing literature in this area focuses on methodological advancements and proof-of-concept studies, rather than developing user-centered machine learning algorithms grounded in real-world data that can outperform current management practices (Meier and Laumer 2022). A common limitation in this field is the lack of access to actual job performance data, leading many studies to rely on proxy measures of person-job fit, such as similarity or distance scores between candidates' credentials and job descriptions (e.g., Malinowski et al., 2006; Kmail et al., 2015; Zaroor et al., 2017; Manad et al., 2018; Bello et al., 2020; Lee and Ahn, 2020; Zhu et al., 2018; Shen et al., 2018; Qin et al., 2020). These studies assume that higher similarity between a candidate's CV and a job description indicates a better fit. Unfortunately, prior studies have not validated this assumption using subsequent job performance data.

[5] This literature is voluminous, and it is beyond the scope of this study to provide a thorough review of this literature. Interested readers can read Strohmeier (2022) and Qin et al. (2024) for comprehensive reviews.

Another stream of research in people analytics involves building prediction models of "hirability," using scores from human raters as ground truth. However, even experienced HR professionals often disagree on the most important factors in candidate evaluation (Harris 2018; Lerner and Bergman 2024). Moreover, these studies are often based on hypothetical scenarios that may not accurately reflect the complexities of real-world hiring decisions. While the management field has begun exploring algorithm-based employee selection, much of the focus remains on the measurement of theoretical constructs rather than prediction (Campion et al. 2024; Sajjadiani, Sojourner, Kammeyer-Mueller, and Mykerezi 2019; Chakraborty, Chiong, Dover, and Sudhir 2024).

Our study differs fundamentally from these existing approaches in several key ways. First, we observe both hiring outcomes and post-hiring performance for all applicants, enabling us to develop more reliable predictive models. This represents a significant advantage over Chen and Ke (2026), who observe post-hiring performance only for applicants who are hired. Second, unlike many studies in the machine learning field that prioritize methodological innovations, we emphasize the critical importance of data engineering (e.g., resume data in our study). Our approach aligns with Ng's (2021) call for a data-centric approach that prioritizes refining input data (X) and labels (Y) over iterative model adjustments.

Although our best machine learning model, Model ABC, demonstrates strong performance, we acknowledge that research productivity is not the sole factor considered by human recruiters. For example, cultural fit may play a significant role in recruitment decisions. Furthermore, many aspects of the recruitment process may not lend themselves to machine learning approaches. Accordingly, we view our machine learning models as tools to augment—not replace—human recruiters by improving the efficiency and quality of candidate screening at the initial stage of the recruitment process. Furthermore, by generating a quantitative estimate of research potential, our models offer an explicit quantitative benchmark that can inform trade-offs among candidate attributes (e.g., research versus teaching potential), thereby potentially enhancing the quality of overall recruitment decisions.

3. Empirical Designs

3.1. Data sources

The data for our study is drawn from multiple sources. To minimize data collection costs, we focus on rookie accounting job candidates who attended the Miami Rookie Camp. Given the need to evaluate each candidate's post-graduation performance over a five-year period, our sample is restricted to rookie candidates from the years 2015 to 2018. For instance, a job candidate from 2015 would have submitted their job application to the Camp by December 31, 2015, and graduated by the summer of 2016. For each candidate, we collect only their resume and job market paper, as the primary objective of this study is to evaluate the predictive value of these data sources for future research productivity. As shown in Table 1, our final sample comprises 569 rookie candidates: 152 from 2015, 141 from 2016, 145 from 2017, and 131 from 2018.

To measure the academic quality of the universities represented in our sample, we gather historical business school rankings from U.S. News & World Report and the QS World University Rankings. The U.S. News rankings, published annually in the Best Graduate Schools report, list the top 50 business schools in the United States. The QS World University Rankings, also published annually, offer a broader global perspective. Since the U.S. News rankings are limited to U.S. institutions and the majority of our candidates received their PhDs from North American universities, we use them as the primary source. For institutions not included in the U.S. News list, we supplement with the QS rankings. Combining these sources, we compile a comprehensive list of 80 ranked universities. To ensure the integrity of our analysis and avoid look-ahead bias, we rely on rankings published in the year preceding each candidate's application. This approach ensures that the rankings used for university reputation assessment accurately reflect the information available at the time candidates enter the job market.

To assess the reputation of a rookie candidate's advisors and coauthors, we utilize the annual research productivity rankings from the BYU website (https://byuaccounting.net/rankings/main/studydescription.php). These rankings are based on publications in prominent accounting journals, including Accounting Horizons, Accounting, Organizations and Society, Auditing: A Journal of Practice & Theory, Behavioral Research in Accounting, Contemporary Accounting Research, Journal of Accounting and Economics, Journal of Accounting Research, Journal of Information Systems, Journal of Management Accounting Research, Journal of the American Taxation Association, Review of Accounting Studies, and The Accounting Review.

We also track the initial job placement of each rookie candidate using publicly available sources, such as online resumes, LinkedIn profiles, and SSRN. For candidates without publicly available placement information or who took positions outside academia (9 candidates in the 2015 cohort, 8 in 2016, 12 in 2017, and 7 in 2018), we conservatively assume that they obtained positions at the lowest-ranked universities.

We measure research productivity using publication data from 11 top journals in accounting and finance: Journal of Accounting Research, Journal of Accounting and Economics, The Accounting Review, Contemporary Accounting Research, Review of Accounting Studies, Accounting, Organizations and Society, Journal of Finance, Journal of Financial Economics, Review of Financial Studies, Journal of Financial and Quantitative Analysis, and Management Science. We include Management Science in this list because it is a leading journal in management science and a popular publication outlet for accounting researchers.

To identify publications authored by the rookie candidates in our sample, we collect all articles published in these journals from 2005 to 2023 and perform a name-matching process. To ensure comparability among PhD graduates from different years, we limit our analysis to papers published within five years after the candidates' Miami Rookie Camp application year. The five-year horizon is chosen because our dataset includes publication records only through the end of 2023 at the start of this study. In addition, assistant professors in many universities are required to submit their tenure applications by the end of year five.

3.2. Definition of research productivity

In our primary analyses, we measure a job candidate's research productivity by counting the number of publications in the 11 top accounting and finance journals over a five-year period starting from January 1 of a candidate's graduation year. As a robustness check, we also define research productivity using a citation-weighted measure in Section 4.4, in which each publication is weighted by its Google citation count as of July 24, 2024, the date of our data collection.

To simplify the prediction task, we convert the continuous research productivity variable into a dichotomous variable, which equals 1 if a rookie candidate's research productivity ranks within the top K% of all rookie candidates graduating in the same year, and 0 otherwise. We evaluate K% values of 5%, 10%, 20%, and 30%.[6] We limit K% to 30% because, as shown in footnote 6, the cutoff for K=30% is already the lowest positive integer of 1. We denote these dichotomous variables as pub_top_5pct, pub_top_10pct, pub_top_20pct, and pub_top_30pct, corresponding to the respective cutoff levels (see Panel A of Appendix A for the detailed definitions). In addition to reducing forecasting complexity, using multiple dichotomous measures of research productivity better aligns with the needs of individual institutions, which are often more concerned with whether a candidate falls within the top K% of the applicant pool than with the exact number of top-tier journal publications.

3.3. Definitions of model predictors

We introduce the predictors of our research productivity prediction models in this section. See Appendix A for the detailed definitions of the predictors.

3.3.1. Predictors extracted from the resume

Given the unstructured nature of resumes, we use GPT-4 (gpt-4-1106-preview) from OpenAI to extract structured variables. The specific prompts used for this process are presented in Table OA1

[6] The number of publications corresponding to the cutoffs of 5%, 10%, 20%, and 30% are as follows: for the test year 2017, the counts are 4, 3, 2, and 1, respectively; for the test year 2018, the counts are 3, 2, 1, and 1, respectively.

of the Online Appendix. We refer to the resulting set of predictors extracted directly from the resumes as Predictors A. We assess the accuracy of GPT's feature extraction from résumés in the Online Appendix, Table OA2.

As noted in the Introduction, recruiters observe multiple signals during the hiring process. Compared with other sources, such as interview evaluations available only to shortlisted candidates, resume data offer a key advantage: the features we study are largely factual, difficult to manipulate, and often verifiable through third-party sources. Models based on resume data are therefore likely more robust to candidates' behavioral responses.

3.3.2. Predictors extracted from external public knowledge

Many predictors extracted from resumes carry significant meaning that is readily apparent to human recruiters due to their external knowledge, but may not be obvious to machines. Hence, we develop the following set of Predictors B for each candidate. First, we measure the academic quality of candidates' institutions using the U.S. News and QS rankings discussed earlier. Second, we assess publication and presentation quality using the top journals and accounting conferences noted in Table OA1. Third, using BYU accounting rankings, we measure the reputation of dissertation advisors and coauthors, as well as the strength of coauthor network. Finally, we infer nationality from languages reported on the resume.

3.3.3. Predictors extracted from the job market paper

Because the job market paper contains unstructured data, we use text embeddings to represent its content numerically. An embedding is a sequence of numbers that captures the semantic content of the text, such as concepts expressed in natural language. Given the study's primary focus on evaluating the predictive value of resumes—and the token constraints of the large language model employed—we restrict the embedded content of each job market paper to its title, abstract, and introduction. For brevity, we continue to refer to this subset of text as the "job market paper" throughout the paper. A comprehensive exploration of full-text embeddings is left to future research.

We use OpenAI's text-embedding-3-large model to generate paper embeddings. Although the model produces 3,072-dimensional vectors, we retain only the first 256 dimensions, which OpenAI reports still perform well relative to most alternatives. For texts longer than 8,191 tokens, we use overlapping sliding windows with a 100-token stride and average the resulting vectors (Goodfellow, Bengio, and Courville 2016). For candidates without job market papers, we assign a 256-dimensional zero vector. We label these measures Predictors C.

Given the limitations of the embedding techniques employed by current large language models, it is unlikely that embeddings will fully capture the informational content of the job market paper's title, abstract, and introduction. This constraint should be considered when interpreting the incremental value of job market papers in our subsequent analysis.

3.4. Machine learning methods

3.4.1. Model details

Our primary objective is to leverage a job candidate's resume and other publicly available data from year t to predict their research productivity over the subsequent five years. Given the temporal nature of our machine learning approach, we adopt a cumulative time series framework, incorporating all available historical data on rookie job candidates to train the models.

Due to the relatively small sample size in each year's rookie job market, we require at least 2 years of training data to ensure model robustness. Consequently, our analysis includes two test years: 2017 and 2018. For the 2017 test year, we use training data from 2015 and 2016. Similarly, for the 2018 test year, we train the model using data from 2015 to 2017.

Prior accounting research often employs advanced machine learning methods for predictive tasks. However, in this study, we use standard logistic regression as our primary method due to the small training sample sizes. Untabulated results indicate that prediction performance using the balanced random forest method is consistently inferior to that of the logistic regression approach.

To evaluate the predictive value of Predictors A, B, and C individually, we train three separate sets of predictive models, each using only one predictor type. Additionally, we assess the predictive value of combining all three predictor types (denoted as Model ABC).

There are two possible approaches for training models that combine predictors A, B, and C. The most common method, widely used in prior studies, involves combining all predictors into a single model for training, known as the early-fusion approach. Alternatively, the late-fusion approach trains separate models for each predictor type and then combines their predictions to produce the final output.

Each approach has distinct advantages and limitations, which depend on the task context, data characteristics, and modeling objectives (Pereira, Salazar, and Vergara 2023). The early-fusion approach allows the model to access all available data upfront, potentially improving performance if the features are complementary. It also enables the model to capture dependencies between features across different data sources. However, early fusion may face challenges when predictors from different modalities have varying scales or distributions, and it is more susceptible to overfitting, particularly with limited training data.

In contrast, the late-fusion approach offers greater flexibility by processing each data source independently, enabling a modular design that is easier to update. This approach is robust to missing data, as predictions can still be made using the remaining models when one data source is unavailable. Additionally, late fusion is more scalable, allowing new data sources or models to be incorporated without requiring retraining of the entire system. It is also less prone to overfitting because individual models operate on smaller, independent feature sets.

In this study, we implement a late-fusion ensemble approach that averages the predicted probabilities from individual models (Baltrušaitis, Ahuja, and Morency 2019). Untabulated results indicate that the early-fusion specification yields inferior predictive performance.

One empirical challenge to estimating our machine learning models is that the observed matches between job candidates and placement institutions are endogenously determined, raising concerns about potential selection biases. To address this, we follow Swaminathan and Joachims (2015) and Schnabel et al. (2016) by implementing an inverse probability weighting (IPW) procedure, which re-weights the training sample to approximate a hypothetical market in which, conditional on observed resume characteristics, candidates are quasi-randomly assigned to different types of institutions. We provide a detailed explanation of this procedure in Appendix B.

To mitigate the risk of overfitting in out-of-sample predictions, we partition the data into training, validation, and test sets, as illustrated in Figure 1. The test set is entirely disjoint from the combined training and validation sample. We randomly allocate 20% of the combined training and validation sample to the validation set, with the remaining 80% used for training. During model training, we specify a hyperparameter grid for C (the inverse of the regularization strength) with values [0.001, 0.01, 0.1, 1]. The optimal C for each model is selected based on the highest ROC_AUC score on the validation set. All other hyperparameters are set to their default values.[7] Because the test set is used only once—after model training and validation are complete—we reduce the likelihood of overfitting.[8]

3.4.2. Performance evaluation metrics

Because the dependent variables for research productivity are binary and highly skewed, we use NDCG@K (normalized discounted cumulative gain at K) as our primary evaluation metric, following Bao et al. (2020). We set K equal to the unconditional mean of the dependent variable in the test sample, which corresponds to the accuracy of a random benchmark. Because NDCG@K is not easily interpretable, we also report Precision@K and Recall@K. Under this choice of K, the two measures are identical, so we focus on Precision@K in the subsequent tables and discussion. Because NDCG@K is not directly comparable across models with different unconditional means of the

---

[7] Specifically, tolerance for stopping criteria: *tol* = 0.0001, maximum number of iterations to converge: *max_iter* = 100. More details of the default parameters of the logit model available at https://scikit-learn.org/1.5/modules/generated/sklearn.linear_model.LogisticRegression.html.

[8] A common approach to assess overfitting risk is to compare predictive performance across in-sample and out-of-sample data (Roelofs 2019; Webster, Rabin, Simon, and Jurie 2019; Aburass and Rumman 2024). For our best-performing model, ABC, untabulated results show that NDCG@K for training-sample is not consistently higher than for out-of-sample predictions. This pattern provides no evidence of systematic overfitting.

dependent variable, we also report LIFT, defined as Precision@K divided by that unconditional mean. Detailed definitions of all metrics are provided in Appendix C.

4. Empirical results

4.1. Descriptive statistics

Table 2 presents the descriptive statistics for all dependent variables and the variables included in Predictors A and B for the full sample spanning 2015–2018. We omit the predictors C as they are text embeddings and difficult to interpret. Panel A highlights that the dependent variables are highly unbalanced. For the four academic performance proxies, the mean values exceed the cut-off thresholds. This imbalance is largely due to ties in publication counts. For instance, while the top 30% of rookies are defined as those who have published at least one paper in top journals, the mean value of *pub_top_30pct* is 0.41.

4.2. Machine learning models for research productivity

Table 3 presents the performance of our machine learning models for the test years 2017 (Panel A) and 2018 (Panel B), respectively. Focusing on the test year 2017 in Panel A, we can draw several insights from the results. First, the individual predictors A, B, and C demonstrate significant value in forecasting rookie candidates' future research productivity when used separately. The LIFT values indicate that, compared with the unconditional mean of the dependent variable, which serves as the benchmark, all models outperform this baseline by a substantial margin. For example, using *pub_top_5pct* as the outcome variable, the Precision@K metric for the model based on Predictor A (Model A) is 7.16 times the benchmark, 8.95 for Predictor B (Model B), and 3.58 for Predictor C (Model

C). Second, the combined model incorporating all predictors (Model ABC) outperforms all individual models A, B, or C when evaluated using NDCG@K as the performance metric.

The conclusions for test year 2018 reported in Panel B are broadly similar, with a few minor differences. Using NDCG@K as the evaluation metric, Model ABC outperforms the best-performing individual model only when K=5%. In contrast, when Precision@K is used as the evaluation metric, Model ABC performs at least as well as the best-performing individual model across specifications.

Focusing on the most comprehensive model, Model ABC, we next evaluate its performance across different cutoffs. Focusing on the test year 2017, LIFT shows a consistent downward trend: 12.53 for K=5%, 3.70 for K=10%, 2.26 for K=20%, and 1.61 for K=30%. We can draw similar inferences for the test year 2018. This monotonic decline highlights that the predictive advantage of Model ABC diminishes as K increases. These findings reflect the nature of the prediction task: as the cutoff K increases, the task becomes inherently easier, reducing the relative benefit provided by the machine learning model. In other words, while Model ABC excels in scenarios where precise predictions are more challenging (e.g., at smaller values of K), its incremental value decreases as the prediction requirements become less stringent.

4.3. Comparing Model ABC with human recruiters

Up to this point, we have used the unconditional mean of the dependent variable, research productivity, as a benchmark for evaluating the performance of our prediction models. A natural question arises: can our best-performing model, ABC, outperform the selection decisions of human recruiters? The answer to this question is not immediately clear due to several factors. On one hand, human recruiters have access to a wealth of additional information beyond what is incorporated into Model ABC. This includes qualitative and "soft" information derived from reference letters and interviews (both online and offline) in the later stages of the recruitment funnel, as well as other context-specific insights that are challenging to quantify. These factors could give human decision-makers a distinct advantage over the algorithmic model.

On the other hand, human recruiters may face significant cognitive limitations that could hinder their ability to effectively process and synthesize vast amounts of information, especially at the initial stage of the recruitment funnel. Furthermore, recruiters' decision-making is not immune to biases and inefficiencies. Agency conflicts, stereotypes, and personal preferences may further distort recruitment decisions, leading to suboptimal candidate selection. These limitations suggest that, despite access to richer information at all stages of the recruitment funnel, human recruiters may not make better selection decisions than Model ABC.

Ideally, evaluating this question would compare the performance of Model ABC to the selection decisions made by recruiters at each individual institution. However, because data on the full applicant pool and placement decisions at the school level are unavailable, we conduct the analysis at the aggregate level—comparing Model ABC's performance to the collective placement outcomes across all institutions. If the rookie job market functions efficiently—such that more productive candidates are matched with higher-ranked institutions, and research potential is the primary selection criterion—then institutional placement rankings should provide a stronger predictor of subsequent research productivity than Model ABC.

4.3.1. The out-of-sample performance results of the Human Selection Model

To test this prediction, we construct a new machine-learning model for research productivity that uses the ranking of the initial job placement as its sole predictor (the Human Selection Model). This model serves as a proxy for the prediction performance of human recruiters in the accounting profession. For university rankings, we rely on the U.S. News & World Report rankings of the top 50 U.S. business schools. For schools not covered by these rankings, we use the QS World University Rankings. Any remaining initial placements not included in either list are assigned the lowest rank.

This continuous ranking-based predictor has potential measurement errors. For instance, a school ranked No. 15 may not differ significantly from a school ranked No. 20. Additionally, candidates who receive offers from both No. 15 and No. 20 schools might choose the slightly lower-ranked No. 20 school for personal reasons, such as spouse or family preferences. To address this concern, we convert the continuous ranking variable into deciles for schools included in the combined U.S. News and QS World University rankings. Schools excluded from both lists are assigned a separate, lowest-ranking decile. [9]

Table 4 compares the performance of Model ABC with the Human Selection Model for the test years 2017 (Panel A) and 2018 (Panel B), respectively. Note that the performance metrics for Model ABC in Table 4 are reproduced from Table 3. As the findings for 2017 and 2018 are broadly similar, our discussion focuses on the results for the test year 2017.

---

[9] The inferences remain consistent regardless of whether the continuous or quintile ranking of initial job placement is applied in the Human Selection Model (results not tabulated).

The analysis of the test year 2017 reveals several key insights. First, the LIFT values indicate that the Precision@K values for the Human Selection Model are consistently higher than the unconditional mean across all K cutoff levels, indicating that human recruiters outperform the baseline. Second, and more importantly, Model ABC consistently outperforms the Human Selection Model at all K cutoff levels, as reflected in its higher NDCG@K values. For example, at K=5%, Model ABC achieves an NDCG@K that is 171% higher than that of the Human Selection Model; at K=10%, the corresponding improvement is 108%. These results highlight the superior predictive accuracy of Model ABC.

However, the relative advantage of Model ABC over the Human Selection Model decreases as the cutoff K increases, with performance differences diminishing from K=5% to K=30%. This pattern suggests that as the prediction task becomes less challenging to human recruiters with larger cutoffs, the benefit of using Model ABC over the Human Selection Model diminishes.

4.3.2. Performance of top-ranked schools

One potential critique of the analysis in Section 4.3.1 is that comparing Model ABC to human recruiters for the entire profession may not be entirely appropriate, as lower-tier schools may lack access to the full pool of rookie candidates. Additionally, top-tier schools may possess superior talent-selection capabilities and thus could beat Model ABC. To address this concern, we compare the performance of Model ABC with the performance of the Human Selection Model, focusing exclusively on top-ranked schools, which are assumed to have full access to the entire pool of rookie candidates.

For each cutoff K (e.g., 5%) in a given test year, we begin by counting the number of rookie candidates (denoted as N) with pub_top_Kpct = 1. Next, we select N rookie candidates whose initial job placements are at the highest-ranked schools, based on the combined U.S. News & World Report and QS World University ranking lists. Using these candidates' subsequent top-journal publications, we calculate relevant performance metrics based on the ranking order of their placement institutions.

Table 5 presents the comparative performance of Model ABC and the Human Selection Model for top-ranked schools in test years 2017 (Panel A) and 2018 (Panel B), respectively. It is important to note that the performance metrics for Model ABC in Panel B are the same as those reported in Table 3. Assuming the top-ranked schools have access to the entire pool of rookie applicants, the unconditional means are the same as in Table 4.

The broad inference in Table 5 is similar to that in Table 4. The analysis reveals little evidence that the selection performance of top-ranked schools exceeds that of Model ABC, especially at lower K cutoffs. This suggests that the predictive advantage of Model ABC remains robust even when compared specifically to top-ranked institutions.

4.3.3. Removing top-ranked schools

The analyses in Sections 4.3.1 and 4.3.2 assume that research potential is the primary selection criterion for all schools. However, teaching could be a critical factor in faculty recruitment at institutions with top MBA programs. To address this potential concern, we exclude candidates placed in the top 10 business schools (as ranked by U.S. News & World Report) and retrain the Human Selection Model on this restricted sample. This adjustment reduces the sample size by 23 candidates across the two test years.

Table 6 presents the test results for the revised Human Selection Model and the original Model ABC for the test years 2017 (Panel A) and 2018 (Panel B), respectively. As we have removed the candidates placed in the top 10 schools, the unconditional means are different from those in Table 4 too. In addition, Model ABC is the same as in Table 4, except that we compute all performance evaluation metrics using the reduced test sample.

The broad inference in Table 6 is similar to that in Table 4. The results show no evidence of improved performance of the Human Selection Model with the revised sample compared to the full sample. This outcome suggests that the weaker performance of the Human Selection Model is unlikely to be due to top school recruiters placing greater weight on teaching than on research potential in recruitment.

4.4. Citation-weighted research productivity

In our primary analysis, we measure a candidate's research productivity by counting top-tier journal publications. However, this measure has a limitation—it does not account for the impact of the publications. To address this concern, we adjust the publication count for impact by weighing each candidate i's publications based on their Google Scholar citation counts measured as of July 24, 2024. The weighted publication count is computed using the formula:

$$Num\_Pub\_Weighted_i = Num\_Pub_i * \frac{Citation_i}{\sum_{k=1}^{\#rookies\ of\ the\ same\ batch} Citation_k}$$

Where Num_Pub_i is the number of top journal publications for candidate i, and $Citation_i$ denotes the number of citations for all top journal publications by candidate i. Using this adjusted performance measure, we redefine the research productivity indicators based on the same percentile cutoffs, denoting them as *pub_w_top_Xpct*, where X=5, 10, 20, or 30, according to $Num\_Pub\_Weighted_i$.[10]

Table 7 presents the test results for the research productivity prediction models using these weighted measures across various K cutoffs for the test years 2017 (Panel A) and 2018 (Panel B). The results show that our inferences remain robust to adjustments for publication quality.

We also replicate the analysis in Table 4 using citation-weighted publications. The results are shown in Table 8 for the test years 2017 (Panel A) and 2018 (Panel B). Again, we find qualitatively similar inferences to those in Table 4. The only notable exception is the test year 2017 with $K = 5\%$, where Model ABC underperforms the Human Selection Model based on NDCG@K, although the two models perform comparably when Precision@K or LIFT is used as the evaluation metric.

4.5. Shapley value analysis

Given that rookie job candidates' resumes and related information, such as school and BYU rankings, are readily accessible to human recruiters, an important question arises: why do recruiters fail to incorporate this value-relevant information effectively into their faculty selection decisions? We investigate this issue by constructing an alternative machine-learning prediction model, referred to as

[10] Because citation counts generally increase over time, publications closer to our citation measurement date (July 24, 2024) are expected to accumulate fewer citations. To account for this, we also construct a measure of research productivity that adjusts citation counts based on each publication's proximity to the measurement date. Our inferences remain qualitatively unchanged (untabulated).

the “Initial Placement Model,” that uses the same predictors (A, B, and C) to predict the school rankings of candidates’ initial placement institutions.

To align with the definitions of the dependent variables of research productivity for Model ABC in Table 3, we convert the continuous initial placement ranking variable into dichotomous variables with the same cutoffs: 5%, 10%, 20%, and 30%. These variables are denoted as *job_top_Xpct*, where X = 5, 10, 20, or 30. Descriptive statistics for these dependent variables are presented in Table 2, while the results for the Initial Placement Models are reported in Table OA3 of the Online Appendix for test years 2017 (Panel A) and 2018 (Panel B), respectively.

To explore how Model ABC and the Initial Placement Model differ in their use of predictors A, B, and C, we use Shapley values (SHAP) to assess the importance of each predictor in each model. Shapley analysis, derived from cooperative game theory, distributes the total gains among players based on their individual contributions, providing insight into the relative importance of each feature in a model’s predictions.

Because the ensemble method combines the three sets of predictors (A, B, and C) in both models, we first compute the Shapley values for each feature in the individual base models and then aggregate these values to compare and rank feature importance (Lundberg et al. 2020; Abdullakutty and Naseem 2024). To ensure comparability across different predictor sets, we normalize the Shapley values within each model so that the sum of their absolute values equals 1, following the methodology of Leinonen et al. (2023). Specifically, for each test year, we calculate the Shapley values for all predictors, sum their absolute values within the predictor set, and divide each individual Shapley value by this total. This normalization ensures that the sum of the absolute Shapley values for each base model equals 1, allowing a fair comparison of the relative contributions of predictors across models.

Given the large number of features in the combined predictor list of A, B, and C, we follow the approach of Chen and Ke (2026) by focusing on the top 10 most important features for each model, as determined by their Shapley values. Table 9 presents the results for the test years 2017 (Panel A) and 2018 (Panel B).

4.5.1. The top 10 features of Model ABC

We first focus on the top 10 features of Model ABC for the test year 2017 in Panel A. Several patterns emerge. First, a stable core of predictors appears across the dependent variable specifications, including measures of collaboration breadth (*number of coauthors*), professional membership and reference prominence (e.g., *reference_first*, *reference_high*, *reference_mean*), institutional prestige (*PhD_top*), and early research achievement (*number of awards* and *number of top R&R papers*).

Second, the relative importance of these predictors varies with the cutoff used to define research productivity. For the most selective research productivity cutoffs (top 5% and 10%), early distinction signals—particularly awards and high-quality R&R outcomes—rank most prominently. In contrast, for broader research productivity definitions (top 20% and 30%), collaboration-based measures dominate, with the *number of coauthors* ranked first and additional network characteristics (e.g., *coauthor_high*, *coauthor_top*, *number_of_coauthor_2degree*) entering the top ranks.

Third, the estimated signs are generally intuitive: awards, coauthorship breadth, R&R quality, and doctoral program prestige are positively associated with future productivity. However, consistent with the model's multivariate nature, several predictors exhibit negative associations that likely reflect conditional trade-offs rather than unconditional effects. The coefficient on *the number of memberships* is negative across specifications, suggesting that, holding other characteristics constant, extensive committee or association involvement may proxy for time allocated away from research activities. Similarly, reference-based prominence measures (e.g., *reference_high* and *reference_first*) enter with negative signs, suggesting that stronger reference reputations are negatively correlated with future research productivity. Finally, at higher productivity cutoffs, measures of coauthor reputation (e.g., *coauthor_high*, *coauthor_top*, and *coauthor_mean*) are negatively associated with future productivity. These interpretations should be viewed with caution, as Shapley-based importance measures conditional predictive contributions rather than structural causal effects.

The 2018 Shapley value results for Model ABC reinforce the main findings from 2017. Extreme productivity is most closely linked to early quality and citation signals, whereas broader upper-tier productivity is more strongly associated with the breadth of collaboration and professional embeddedness. There exists modest year-to-year variation in the relative ranking of these dimensions. In addition, the signs of key variables are highly stable across the two test years. Awards, coauthorship breadth, R&R-based measures, doctoral program prestige, and reviewer experience remain positively associated with future productivity, while *the number of memberships* and several conditional reputation measures (e.g., *reference_first* and *reference_high*) continue to have negative marginal contributions. These patterns suggest that the cross-year differences primarily reflect reweighting among closely related predictors rather than a structural change in the model's economic interpretation.

4.5.2. The top 10 features of the Initial Placement Model

We now turn to the top 10 features of the initial placement models. We start with an analysis of the test year 2017. Across all cutoff definitions (5%, 10%, 20%, and 30%), several predictors consistently rank among the most important features and can be grouped into four broad economic categories. First, measures of research productivity and visibility—most notably, the number of awards and *presentations*—appear in all four models and are uniformly positively associated with top placements. Presentations at top conferences (*number of presentations at top conferences*) also enter prominently in the 10% and 20% specifications with positive signs. These patterns suggest that external validation and professional visibility are strongly associated with favorable placement outcomes across the distribution, consistent with the view that hiring committees reward signals of research momentum and peer recognition.

Second, advisor and reference quality variables appear in most specifications. Measures such as *reference_first*, *reference_mean*, and *reference_high* enter three or more models, although their signs are not always stable across cutoffs. Their repeated inclusion nevertheless indicates that the reference writer's reputation is an important predictor of initial placement, particularly at more selective thresholds.

Third, network and coauthor characteristics frequently emerge as important predictors. Variables such as *coauthor_high*, *number_of_coauthor_2degree*, and *number of membership* appear across multiple cutoffs. While their estimated signs vary in some cases, the number of memberships is consistently negative in the 10%, 20%, and 30% models, suggesting that broader but potentially weaker affiliations may not translate into stronger placement outcomes.

Finally, educational background—captured by *PhD_top*—enters the 5% and 20% specifications with a positive sign. This finding is consistent with institutional prestige serving as a salient signal in the academic job market.

Taken together, the Shapley value results for test year 2017 indicate that research productivity, reference quality, network characteristics, and doctoral program prestige jointly shape initial placement outcomes, with productivity-related measures exhibiting the most consistent positive associations across cutoff definitions.

We next move to the Shapley value results for the initial placement models for the test year 2018. We find that both test years share a common structure: research productivity, reference quality, network characteristics, and doctoral prestige jointly shape initial placement outcomes. Although the relative importance of specific variables shifts across years and cutoffs, the underlying economic categories that drive placements remain remarkably stable.

4.5.3. Differences between Model ABC and the Initial Placement Model

Turning to a comparison of the top 10 features in the Initial Placement Model and Model ABC, several observations emerge. Although the two models share several highly ranked features, their feature compositions differ substantially, particularly at lower cutoff levels. For example, visibility-related variables—such as *number of presentations* and *number of presentations at top conferences*—and specialization in financial accounting (*primaryResearchArea_financial*) appear more frequently among the top 10 features in the Initial Placement Model. In contrast, Model ABC places greater weight on measures of collaboration breadth (e.g., coauthors), reference prominence (e.g., *reference_first*, *reference_high*, *reference_mean*), institutional prestige (*PhD_top*), and early research achievement (e.g., *number of R&R papers* and *number of top R&R papers*).

Second, the direction of feature importance often diverges between the two models, particularly at lower cutoffs such as K=5% and 10%. For instance, in test year 2017, while Model ABC finds that advisors' reputation (reference_high) and coauthors' mean reputation (coauthor_mean) negatively predict a rookie candidate's future research productivity, the Initial Placement Model assigns positive weights to these factors. Conversely, Model ABC views external visibility (*number of presentations*) listed on a candidate's resume as a negative predictor of future productivity, whereas the Initial Placement Model assigns a positive weight to this factor.

Even when the directions of feature importance align between the two models, the feature weightings differ significantly. For example, although the *number of top R&R papers* contributes positively in both models, it appears among the top 10 features only in Model ABC. Given that Model ABC outperforms human recruiters at predicting research productivity, these differences in Shapley values suggest that human recruiters misallocate weights among predictors, which likely contributes to their relatively weaker performance.

4.6. Implications of Model ABC for the accounting profession's rookie talent distribution

Given that machine learning has not yet been adopted in rookie faculty recruitment, it is informative to explore the potential impact of implementing our best-performing model, Model ABC, on the allocation of accounting rookie talent across institutions. We conduct a hypothetical analysis under two simplifying assumptions: (i) recruiters base hiring decisions solely on candidates' predicted research potential, and (ii) rookie candidates always accept offers from the highest-ranked institution, where institutional rankings are determined by a combination of U.S. News & World Report and QS World University Rankings.

We evaluate the impact of Model ABC in two steps. First, we compute the market share of top-tier journal publications over the five-year post-graduation period—both unadjusted and citation-adjusted—for the rookies actually hired by the top 10 U.S. schools in each test year (denoted as SHARE_UNADJ and SHARE_ADJ, respectively). In both 2017 and 2018, these schools hired 10 candidates, representing less than 5% of the rookie market in each year. We refer to this group as the "Human Hired Group." Second, we identify the same number of candidates using Model ABC's predicted pub_top_5pct scores and compute the same market-share measures for this group, denoted as the "ML Selected Group." We use *pub_top_5pct* to match the actual hiring proportion of the top 10 schools in both years.

Table 10 reports the top 10 schools' market share of top-tier publications—measured by SHARE_UNADJ and SHARE_ADJ—for both the Human Hired Group and the ML Selected Group. Based on SHARE_UNADJ, the 10 rookies actually hired by the top 10 schools account for 17.19% and 22.77% of all top journal publications in the five years following graduation for the 2017 and 2018 cohorts, respectively. In contrast, had these schools hired the top 10 candidates as ranked by Model ABC, their market share would have risen to 31.25% and 25.74%, reflecting economically meaningful increases of 82% and 13%, respectively. Results using the citation-adjusted measure, SHARE_ADJ, yield qualitatively similar conclusions. Taken together, these findings highlight the potential for machine learning–based hiring models to significantly alter the distribution of academic talent across institutions.

5. Are there systematic biases in human recruiters' selection outcomes?

In this section, we examine whether human recruiters' suboptimal weighting of predictors A, B, and C represents systematic biases against specific groups of candidates. To investigate this, we develop a metric, denoted as *gap_Kpct_prob*, which quantifies the discrepancy between predictions generated by Model ABC and those produced by the Initial Placement Model at each cutoff K. Specifically, for each test year, we calculate $prob(pub\_top\_Kpct_{ABC})$, representing the predicted probabilities of *pub_top_Kpct* for Model ABC. Similarly, we compute $prob(job\_top\_Kpct_{ABC})$ for the Initial Placement Model, which represents the predicted probabilities of *job_top_Kpct* for the initial placement outcomes. *gap_Kpct_prob* is defined as follows:

$$gap\_Kpct\_prob = prob(pub\_top\_Kpct_{ABC}) - prob(job\_top\_Kpct_{ABC})$$

Since both Model ABC and the Initial Placement Model rely solely on information available at the time of rookie faculty recruitment and are evaluated out of sample, *gap_Kpct_prob* serves as a reliable proxy for the potential difference between the optimal outcomes (as predicted by Model ABC) and the actual outcomes chosen by human recruiters (as represented by the Initial Placement Model). A *gap_Kpct_prob* value of 0 indicates that Model ABC's prediction perfectly aligns with that of the Initial Placement Model. A positive value suggests that human recruiters underestimate a candidate's research productivity relative to the optimal one, while a negative value indicates that human recruiters overestimate a candidate's research productivity relative to the optimal one.

We then investigate whether *gap_Kpct_prob* varies systematically with the following two categories of candidate characteristics: (i) demographic characteristics, including *gender*, whether the candidate is Asian (second_language_asia), whether the candidate is European (*second_language_euro*); and (ii) institution background, including whether the candidate earned a PhD from a top-10 business school as ranked by U.S. News (*PhD_top10*), and whether the candidate's primary dissertation advisor ranks in the top 10% of the BYU ranking list for accounting journal publications in the test year (*reference_top*).

Table 11 reports the regression results. Panel A presents descriptive statistics for *gap_Kpct_prob* across alternative cutoff values of K. The mean *gap_Kpct_prob* is positive when K=5% and 10% but negative when K=20% and 30%. This pattern suggests that, on average, the human recruiters' Initial Placement Model underestimates the future research productivity of top-performing rookies and overestimates the productivity of moderately performing rookies.

Panel B reports the OLS estimates for *gap_Kpct_prob*, where all candidate characteristics are included in a single specification. Regarding demographic characteristics, we find that the coefficients on *gender* and *second_language_asia* are generally statistically insignificant across cutoffs, providing no evidence of systematic human bias against female candidates or candidates with an Asian language background. In contrast, the coefficient on *second_language_euro* is negative at all cutoff values and statistically significant for K=20% and 30%. This pattern suggests that recruiters tend to overestimate the future research productivity of candidates with a European language background at higher cutoff levels.

Regarding institutional background, we find that the coefficient on *PhD_top10* is negative across all cutoffs and statistically significant for K=10% and 20%, while the coefficient on *reference_top* is negative at all cutoffs and significant for K=10%, 20%, and 30%. Taken together, these findings are consistent with recruiters systematically overestimating the expected productivity of candidates from elite PhD programs and those endorsed by leading accounting scholars during the initial hiring stage.

6. Conclusion

This study examines whether limited information available at the initial stage of the employee selection process can be used to develop an effective machine learning model for candidate screening. We focus on accounting rookies, whose positions require substantial discretion and sustained research productivity, and evaluate whether résumés, publicly available metrics on academic institutions, PhD advisors, and coauthors, and job market papers predict long-term research output.

We find that résumé-based information alone is a strong predictor of subsequent performance and, in several specifications, outperforms human screening decisions. In addition, the machine learning model exhibits lower bias relative to human recruiters. Taken together, these findings suggest that models based on information readily available early in the recruitment process can improve screening accuracy and mitigate inefficiencies inherent in human-centered hiring.

This research makes several important contributions. First, it highlights the utility of machine learning in addressing a longstanding challenge in recruitment: identifying high-potential candidates from large applicant pools during initial screening with limited candidate information. By focusing on academic hiring—a context that requires high discretion and long-term productivity—we extend the scope of existing research, such as Chen and Ke (2026) on machine learning in recruitment, which has largely been limited to entry-level positions with minimal discretion. Second, our study underscores the predictive power of resumes, a widely accessible and underutilized resource in hiring decisions. By integrating resume data with public metrics, our models achieve significant performance gains, offering a scalable solution for improving recruitment outcomes. Third, we provide a novel comparison of machine learning predictions with human selection decisions, revealing systematic biases and suboptimal weighting of predictors in human judgment that could be overcome by machine learning.

The findings have important practical implications. For academic institutions, machine learning models can complement traditional recruitment processes, reducing biases and inefficiencies while improving the likelihood of long-term success among hires. These models also offer a cost-effective and scalable alternative to resource-intensive interview processes, particularly at the initial stages of screening.

More broadly, organizations in non-academic sectors could adopt similar approaches to streamline recruitment. The demonstrated effectiveness of resumes as predictors suggests that organizations could refine their screening methods without extensive reliance on proprietary or confidential data. However, it is essential to ensure that machine learning models are applied ethically, with attention to potential biases in training data that may perpetuate discrimination.

While our study provides valuable insights, it also raises important questions for future research. First, extending the analysis to other academic disciplines and job types would validate the generalizability of these findings. Future studies could explore roles that require different combinations of technical and interpersonal skills to examine the applicability of resume-based predictions across diverse contexts.

Second, incorporating additional data sources, such as structured evaluations of soft skills or references, could further enhance model performance in roles that emphasize leadership and collaboration. Advances in natural language processing may enable the inclusion of qualitative data, such as interview transcripts or letters of recommendation, into predictive models.

Finally, longitudinal studies are needed to examine the broader organizational impacts of integrating machine learning into talent recruitment. These studies should consider not only the predictive accuracy of models but also their effects on workforce diversity, retention rates, and overall productivity. Addressing these questions will be critical for designing recruitment processes that are both effective and equitable.

References

Abdullakutty, F., and U. Naseem. 2024. Decoding memes: a comprehensive analysis of late and early fusion models for explainable meme analysis. In *Companion Proceedings of the ACM Web Conference 2024*, 1681–1689. Singapore: ACM.

Aburass, S., and M. A. Rumman. 2024. Quantifying overfitting: introducing the overfitting index. In *2024 International Conference on Electrical, Computer and Energy Technologies*, 1–7.

Austin, P. C., and E. A. Stuart. 2015. Moving towards best practice when using inverse probability of treatment weighting (IPTW) using the propensity score to estimate causal treatment effects in observational studies. *Statistics in Medicine* 34 (28): 3661–3679.

Baltrušaitis, T., C. Ahuja, and L.-P. Morency. 2019. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41 (2): 423–443.

Bao, Y., B. Ke, B. Li, Y. J. Yu, and J. Zhang. 2020. Detecting accounting fraud in publicly traded us firms using a machine learning approach. *Journal of Accounting Research* 58 (1): 199–235.

Bello, M., L. Alejandra, E. Bonilla, C. Hernandez, B. Pedroza, and A. Portilla. 2020. A novel profile's selection algorithm using AI. *Applied Computer Science* 16 (1): 18–32.

Boushey, H., and S. J. Glynn. 2012. There are significant business costs to replacing employees. *Center for American Progress* 16: 1–9.

Campion, E. D., M. A. Campion, J. Johnson, T. R. Carretta, S. Romay, B. Dirr, A. Deregla, and A. Mouton. 2024. Using natural language processing to increase prediction and reduce subgroup differences in personnel selection decisions. *Journal of Applied Psychology* 109 (3): 307.

Casas-Arce, P., M. M. Cheng, I. Grabner, and S. Modell. 2022. Managerial accounting for decision-making and planning. *Journal of Management Accounting Research* 34 (1): 1–7.

Chakraborty, I., K. Chiong, H. Dover, and K. Sudhir. 2024. Can AI and AI-hybrids detect persuasion skills? Salesforce hiring with conversational video interviews. *Marketing Science* 44 (1): 30–53.

Chen, C., and B. Ke. 2026. The Predictive Value of Job Application Data in Initial Applicant Screening. *Working paper*.

Conley, J. P., and A. S. Önder. 2014. The research productivity of new PhDs in economics: The surprisingly high non-success of the successful. *Journal of Economic Perspectives* 28 (3): 205–216.

Crump, R. K., V. J. Hotz, G. W. Imbens, and O. A. Mitnik. 2009. Dealing with limited overlap in estimation of average treatment effects. *Biometrika* 96 (1): 187–199.

Defazio, D., A. Lockett, and M. Wright. 2009. Funding incentives, collaborative dynamics and scientific productivity: Evidence from the EU framework program. *Research Policy* 38 (2): 293–305.

Deller, C. 2023. Beyond performance: does assessed potential matter to employees' voluntary departure decisions? *Journal of Accounting Research* 61 (4): 981–1024.

Ding, K., B. Lev, X. Peng, T. Sun, and M. A. Vasarhelyi. 2020. Machine learning improves accounting estimates: evidence from insurance payments. *Review of Accounting Studies* 25 (3): 1098–1134.

Ductor, L. 2015. Does co-authorship lead to higher academic productivity? *Oxford Bulletin of Economics and Statistics* 77 (3): 385–407.

Ductor, L., M. Fafchamps, S. Goyal, and M. J. van der Leij. 2014. Social networks and research output. *Review of Economics and Statistics* 96 (5): 936–948.

Ductor, L., and B. Visser. 2022. When a coauthor joins an editorial board. *Journal of Economic Behavior & Organization* 200: 576–595.

Goodfellow, I., Y. Bengio, and A. Courville. 2016. *Deep learning*. Cambridge, Massachusetts: MIT Press.

Grabner, I., J. Künneke, and F. Moers. 2025. Promotion decisions and the adoption of explicit potential assessment. *Management Science* 71 (11): 9233–9255.

Gulbrandsen, M., and J.-C. Smeby. 2005. Industry funding and university professors' research performance. *Research Policy* 34 (6): 932–950.

Harris, C. G. 2018. Making better job hiring decisions using "human in the loop" techniques. In *Proceedings of the 2nd International Workshop on Augmenting Intelligence with Humans-in-the-Loop*, 16–26.

Hernan, M. A., and J. M. Robins. 2020. *Causal Inference: What If*. CRC Press.

Hilmer, M., and C. E. Hilmer. 2012. On the search for talent in academic labor markets: Easily observable later-graduate study outcomes as predictors of early-career publishing, placement, and tenure. *Economic Inquiry* 50 (1): 232–247.

Hoffman, M., L. Kahn, and D. Li. 2018. Discretion in hiring. *Quarterly Journal of Economics* 133 (2): 765–800.

Hussey, A., S. Murray, and W. Stock. 2022. Gender, coauthorship, and academic outcomes in economics. *Economic Inquiry* 60 (2): 465–484.

Kahneman, D., O. Sibony, and C. R. Sunstein. 2021. *Noise: A flaw in human judgment*. New York, NY, USA: Little, Brown Spark.

Kaplan, R. S. 1998. Creating new management practice through innovation action research. SSRN Scholarly Paper. Rochester, NY: Social Science Research Network.

Kmail, A. B., M. Maree, M. Belkhatir, and S. M. Alhashmi. 2015. An automatic online recruitment system based on exploiting multiple semantic resources and concept-relatedness measures. In *2015 IEEE 27th International Conference on Tools with Artificial Intelligence (ICTAI)*, 620–627.

Krapf, M., H. W. Ursprung, and C. Zimmermann. 2017. Parenthood and productivity of highly skilled labor: Evidence from the groves of academe. *Journal of Economic Behavior & Organization* 140: 147–175.

Labro, E. 2025. Thoughts on management accounting research at The Accounting Review's centennial. *The Accounting Review* 100 (6): 373–384.

Lee, D., and C. Ahn. 2020. Industrial human resource management optimization based on skills and characteristics. *Computers & Industrial Engineering* 144: 106463.

Leinonen, J., U. Hamann, I. Sideris, and U. Germann. 2023. Thunderstorm nowcasting with deep learning: A multi-hazard data fusion model. *Geophysical Research Letters* 50 (8): e2022GL101626.

Lerner, A., and P. Bergman. 2024. Are recruiters better than a coin flip at judging resumes? Here's the data. https://interviewing.io/blog/are-recruiters-better-than-a-coin-flip-at-judging-resumes.

Li, D., L. R. Raymond, and P. Bergman. 2026. Hiring as exploration. *Review of Economic Studies* 93 (2): 1200–1240.

Liu, M. 2022. Assessing Human Information Processing in Lending Decisions: A Machine Learning Approach. *Journal of Accounting Research* 60 (2): 607–651.

Lundberg, S. M., G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee. 2020. From local explanations to global understanding with explainable AI for trees.

Nature machine intelligence 2 (1): 56–67.

Malinowski, J., T. Keim, O. Wendt, and T. Weitzel. 2006. Matching people and jobs: A bilateral recommendation approach. In *Proceedings of the 39th Annual Hawaii International Conference on System Sciences (HICSS'06)*, 6:137c–137c. IEEE.

Manad, O., M. Bentounsi, and P. Darmon. 2018. Enhancing talent search by integrating and querying big HR data. In *2018 IEEE International Conference on Big Data (Big Data)*, 4095–4100. IEEE.

Manthei, K., D. Sliwka, and T. Vogelsang. 2023. Information, incentives, and attention: a field experiment on the interaction of management controls. *The Accounting Review* 98 (5): 455–479.

Meier, F. J., and S. Laumer. 2022. Machine learning in HR staffing. In *Handbook of Research on Artificial Intelligence in Human Resource Management*. Edward Elgar.

Merchant, K. A., and W. A. V. der Stede. 2017. *Management Control Systems: Performance Measurement, Evaluation and Incentives*. Pearson.

Mullainathan, S. 2025. Economics in the age of algorithms. *AEA Papers and Proceedings* 115: 1–23.

Ng, A. 2021. A chat with Andrew on MLOps: from model-centric to data-centric AI. https://www.youtube.com/watch?v=06-AZXmwHjo.

Oyer, P. 2006. Initial labor market conditions and long-term outcomes for economists. *Journal of Economic Perspectives* 20 (3): 143–160.

Pereira, L. M., A. Salazar, and L. Vergara. 2023. A comparative analysis of early and late fusion for the multimodal two-class problem. *IEEE Access* 11: 84283–84300.

Qin, C., L. Zhang, Y. Cheng, R. Zha, D. Shen, Q. Zhang, X. Chen, et al. 2024. A comprehensive survey of artificial intelligence techniques for talent analytics. arXiv:2307.03195.

Qin, C., H. Zhu, T. Xu, C. Zhu, C. Ma, E. Chen, and H. Xiong. 2020. An enhanced neural network approach to person-job fit in talent recruitment. *ACM Transactions on Information Systems (TOIS)* 38 (2): 1–33.

Roelofs, R. 2019. Measuring Generalization and Overfitting in Machine Learning. University of California, Berkeley.

Sajjadiani, S., A. J. Sojourner, J. D. Kammeyer-Mueller, and E. Mykerezi. 2019. Using machine learning to translate applicant work history into predictors of performance and turnover. *Journal of Applied Psychology* 104 (10): 1207.

Schnabel, T., A. Swaminathan, A. Singh, N. Chandak, and T. Joachims. 2016. Recommendations as treatments: debiasing learning and evaluation. In *Proceedings of The 33rd International Conference on Machine Learning*, 1670–1679. PMLR.

Shen, D., H. Zhu, C. Zhu, T. Xu, C. Ma, and H. Xiong. 2018. A joint learning approach to intelligent job interview assessment. In *Proceedings of the 27th International Joint Conference on Artificial Intelligence*, 3542–3548. IJCAI'18. Stockholm, Sweden: AAAI Press.

Strohmeier, S. 2022. *Handbook of Research on Artificial Intelligence in Human Resource Management*. Edward Elgar.

Swaminathan, A., and T. Joachims. 2015. Counterfactual risk minimization: learning from logged bandit feedback. In *Proceedings of the 32nd International Conference on Machine Learning*, 814–823. PMLR.

Webster, R., J. Rabin, L. Simon, and F. Jurie. 2019. Detecting overfitting of deep generative networks via latent recovery. In , 11273–11282.

Yadav, A., J. McHale, and S. O'Neill. 2023. How does co-authoring with a star affect scientists' productivity? Evidence from small open economies. *Research Policy* 52 (1): 104660.

Zaroor, A., M. Maree, and M. Sabha. 2017. JRC: A job post and resume classification system for online recruitment. In *2017 IEEE 29th International Conference on Tools with Artificial Intelligence (ICTAI)*, 780–787. IEEE.

Zhu, C., H. Zhu, H. Xiong, C. Ma, F. Xie, P. Ding, and P. Li. 2018. Person-job fit: Adapting the right talent for the right job with joint representation learning. *ACM Transactions on Management Information Systems (TMIS)* 9 (3): 1–17.

Figure 1. Training set, validation set, and test set split

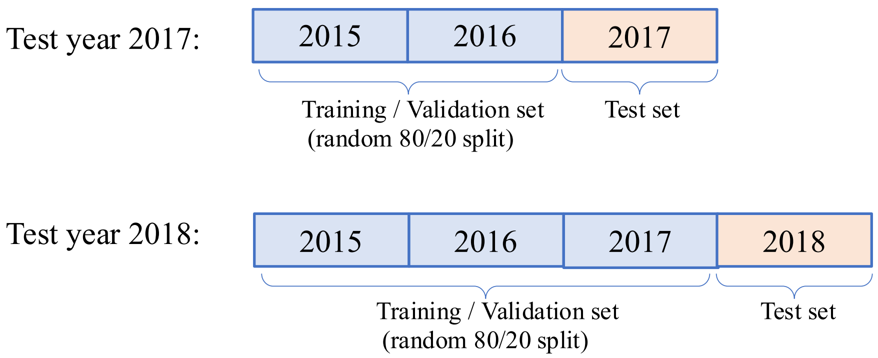

Table 1. Sample selection procedures

original files (number of applicants)	728
remove non-rookie applicants	-53
remove applicants without CV or dissertation	-47
remove duplicated applications of the same person in the same year	-50
remove applications of the same person in different years by keeping the earliest one	-9
final sample	= 569
2015	152
2016	141
2017	145
2018	131

Table 2. Descriptive statistics

	mean	std	min	25%	50%	75%	max
Panel A: Dependent variables							
pub_top_5pct	0.07	0.25	0	0	0	0	1
pub_top_10pct	0.16	0.37	0	0	0	0	1
pub_top_20pct	0.31	0.46	0	0	0	1	1
pub_top_30pct	0.41	0.49	0	0	0	1	1
job_top_5pct	0.05	0.21	0	0	0	0	1
job_top_10pct	0.08	0.28	0	0	0	0	1
job_top_20pct	0.18	0.39	0	0	0	0	1
job_top_30pct	0.30	0.46	0	0	0	1	1
Panel B: Predictors extracted from the resume and public sources (Predictors A and B)							
gender	0.58	0.49	0	0	1	1	1
has Bachelor honor	0.47	0.50	0	0	0	1	1
has Master honor	0.21	0.41	0	0	0	0	1
has PhD honor	0.05	0.22	0	0	0	0	1
number of published papers	0.53	1.02	0	0	0	1	12
number of R&R papers	0.32	0.63	0	0	0	1	5
number of papers in progress	2.74	1.55	0	2	3	4	9
has_coauthor	0.96	0.21	0	1	1	1	1
has_reference	0.83	0.38	0	1	1	1	1
number of coauthors	4.49	2.72	0	3	4	6	18
number of presentations	6.72	4.59	0	4	6	9	30
number of teaching experiences	3.82	2.29	0	2	3	5	13
number of awards	4.48	3.10	0	2	4	6	17
number of reviewers	1.67	1.76	0	0	1	3	11
number of membership	1.23	1.63	0	0	0	2	7
number of working experiences	2.38	1.74	0	1	2	3	11
had academic work	0.65	0.48	0	0	1	1	1
had non-academic work	0.76	0.43	0	1	1	1	1
provide abstract	0.38	0.49	0	0	0	1	1
PrimaryResearchArea_financial	0.64	0.48	0	0	1	1	1
PrimaryResearchArea_auditing	0.15	0.36	0	0	0	0	1
PrimaryResearchArea_managerial	0.12	0.33	0	0	0	0	1
PrimaryResearchArea_tax	0.05	0.23	0	0	0	0	1
PrimaryResearchMethod_archival	0.74	0.44	0	0	1	1	1
PrimaryResearchMethod_experiment	0.15	0.36	0	0	0	0	1
PrimaryResearchMethod_analytical	0.08	0.27	0	0	0	0	1
multi_language	0.19	0.39	0	0	0	0	1
Bachelor_top	0.20	0.40	0	0	0	0	1
Master_top	0.29	0.45	0	0	0	1	1
PhD_top	0.45	0.50	0	0	0	1	1
visit_top	0.07	0.25	0	0	0	0	1
number of top published papers	0.08	0.30	0	0	0	0	2
number of top R&R papers	0.16	0.40	0	0	0	0	2

coauthor_mean	1486.40	1128.96	0	565.20	1386.43	2021	5614
coauthor_high	470.78	939.46	0	26	131	430	5614
number_of_coauthor_2degree	37.51	31.97	0	14	30	51	202
coauthor_top	0.29	0.46	0	0	0	1	1
number of presentations at top conferences	1.56	1.80	0	0	1	2	10
reference_first	651.12	1082.80	0	52	252	708	5614
reference_mean	882.12	864.48	0	252	644.75	1245.20	5614
reference_high	231.17	412	0	19	114	252	5614
second_language_asia	0.11	0.32	0	0	0	0	1
second_language_euro	0.22	0.41	0	0	0	0	1
Placerank	9.05	3.16	1	8	11	11	11

Notes: This table reports descriptive statistics for the dependent variables and the observable predictors extracted from candidates' resumes (Predictors A) and external public sources (Predictors B) for the full sample of rookie candidates from 2015 to 2018. Panel A reports the four research productivity indicators, *pub_top_5pct, pub_top_10pct, pub_top_20pct,* and *pub_top_30pct*, and the four initial placement indicators, *job_top_5pct, job_top_10pct, job_top_20pct,* and *job_top_30pct*. Panel B reports summary statistics for Predictors A and B. Predictors C are omitted because they are text embeddings and are difficult to interpret. See Appendix A for detailed variable definitions.

Table 3. Prediction performance of rookie candidates' research productivity

Panel A. Test year 2017

Predictor	Mean DV	NDCG@K	Precision@K	LIFT
Subpanel A: DV = pub_top_5pct				
A	6.21	59.18	44.44	716.05
B	6.21	61.92	55.56	895.06
C	6.21	22.24	22.22	358.02
ABC	6.21	83.07	77.78	1253.09
Subpanel B: DV = pub_top_10pct				
A	9.66	41.90	28.57	295.92
B	9.66	39.50	28.57	295.92
C	9.66	24.78	28.57	295.92
ABC	9.66	48.20	35.71	369.90
Subpanel C: DV = pub_top_20pct				
A	21.38	39.65	35.48	165.97
B	21.38	53.58	45.16	211.24
C	21.38	46.53	51.61	241.42
ABC	21.38	57.44	48.39	226.33
Subpanel D: DV = pub_top_30pct				
A	43.45	68.59	63.49	146.13
B	43.45	67.12	66.67	153.44
C	43.45	55.12	55.56	127.87
ABC	43.45	73.95	69.84	160.75

Panel B. Test year 2018

Predictor	Mean DV	NDCG@K	Precision@K	LIFT
Subpanel A: DV = pub_top_5pct				
A	6.11	25.29	12.50	204.69
B	6.11	32.95	37.50	614.06
C	6.11	41.25	25.00	409.38
ABC	6.11	57.27	50.00	818.75
Subpanel B: DV = pub_top_10pct				
A	18.32	33.24	29.17	159.20
B	18.32	50.63	37.50	204.69
C	18.32	39.65	33.33	181.94
ABC	18.32	48.94	37.50	204.69
Subpanel C: DV = pub_top_20pct				
A	42.75	43.69	44.64	104.43
B	42.75	66.18	58.93	137.85
C	42.75	50.13	50.00	116.96
ABC	42.75	60.13	58.93	137.85
Subpanel D: DV = pub_top_30pct				

A	42.75	44.73	44.64	104.43
B	42.75	64.30	57.14	133.67
C	42.75	60.48	55.36	129.50
ABC	42.75	59.04	58.93	137.85

Notes: This table reports the out-of-sample predictive performance of models forecasting rookie candidates' future research productivity. Panel A reports results for the test year 2017. Panel B reports results for the test year 2018. Predictors A includes variables extracted directly from candidates' resumes, Predictors B includes variables constructed from external public knowledge, Predictors C includes text embeddings from the job market paper, and ABC is the late-fusion ensemble model that averages the predicted probabilities from the three individual models. Mean DV is the unconditional mean of the dependent variable in the test sample. NDCG@K, Precision@K, and LIFT are defined in Appendix C.

Table 4. Comparing the prediction performance of Model ABC and the Human Selection Model

Panel A. Test year 2017

	pub_top_5pct	pub_top_10pct	pub_top_20pct	pub_top_30pct
Subpanel A: Human Selection Model				
Mean DV	6.21	9.66	21.38	43.45
NDCG@K	30.62	23.21	44.58	57.59
Precision@K	33.33	21.43	41.94	52.38
LIFT	537.04	221.94	196.15	120.56
Subpanel B: Model ABC				
Mean DV	6.21	9.66	21.38	43.45
NDCG@K	83.07	48.20	57.44	73.95
Precision@K	77.78	35.71	48.39	69.84
LIFT	1253.09	369.90	226.33	160.75

Panel B. Test year 2018

	pub_top_5pct	pub_top_10pct	pub_top_20pct	pub_top_30pct
Subpanel A: Human Selection Model				
Mean DV	6.11	18.32	42.75	42.75
NDCG@K	34.18	40.91	64.97	64.97
Precision@K	37.50	29.17	60.71	60.71
LIFT	614.06	159.20	142.03	142.03
Subpanel B: Model ABC				
Mean DV	6.11	18.32	42.75	42.75
NDCG@K	57.27	48.94	60.13	59.04
Precision@K	50.00	37.50	58.93	58.93
LIFT	818.75	204.69	137.85	137.85

Notes: This table compares the out-of-sample prediction performance of Model ABC and the Human Selection Model in forecasting research productivity. The Human Selection Model uses the decile ranking of the candidate's initial placement institution as the sole predictor. Panel A reports test year 2017, and Panel B reports test year 2018. The results for Model ABC are copied from Table 3. Mean DV is the unconditional mean of the dependent variable in the test sample. NDCG@K, Precision@K, and LIFT are defined in Appendix C.

Table 5. Comparing the prediction performance of Model ABC and the Human Selection Model for the top-ranked schools

Panel A. Test year 2017

	pub_top_5pct	pub_top_10pct	pub_top_20pct	pub_top_30pct
	Subpanel A: Prediction performance of the top-ranked schools			
Mean DV	6.21	9.66	21.38	43.45
NDCG@K	27.54	20.88	45.90	62.38
Precision@K	33.33	21.43	41.94	55.56
LIFT	537.04	221.94	196.15	127.87
	Subpanel B: Model ABC			
Mean DV	6.21	9.66	21.38	43.45
NDCG@K	83.07	48.20	57.44	73.95
Precision@K	77.78	35.71	48.39	69.84
LIFT	1253.09	369.90	226.33	160.75

Panel B. Test year 2018

	pub_top_5pct	pub_top_10pct	pub_top_20pct	pub_top_30pct
	Subpanel A: Prediction performance of top-ranked schools			
Mean DV	6.11	18.32	42.75	42.75
NDCG@K	29.69	38.34	65.99	65.99
Precision@K	37.50	29.17	62.50	62.50
LIFT	614.06	159.20	146.21	146.21
	Subpanel B: Model ABC			
Mean DV	6.11	18.32	42.75	42.75
NDCG@K	57.27	48.94	60.13	59.04
Precision@K	50.00	37.50	58.93	58.93
LIFT	818.75	204.69	137.85	137.85

Notes: This table compares the predictive performance of Model ABC against the realized selection outcomes for the top-ranked schools. For each cutoff K in a given test year, we first count the number of candidates with *pub_top_Kpct = 1* and denote that number by N. We then identify the N candidates whose initial placements are at the highest-ranked institutions based on the combined U.S. News and QS rankings, and compute the corresponding performance metrics. Panel A reports test year 2017, and Panel B reports test year 2018. The results for Model ABC are copied from Table 3. Mean DV is the unconditional mean of the dependent variable in the test sample. NDCG@K, Precision@K, and LIFT are defined in Appendix C.

Table 6. Comparing the prediction performance of Model ABC and Human Selection Model after excluding the top 10 business schools

Panel A. Test year 2017

	pub_top_5pct	pub_top_10pct	pub_top_20pct	pub_top_30pct
Subpanel A: Human Selection Model				
Mean DV	4.38	8.03	18.25	40.88
NDCG@K	0.00	14.30	33.02	51.96
Precision@K	0.00	18.18	28.00	51.79
LIFT	0.00	226.45	153.44	126.69
Subpanel B: Model ABC				
Mean DV	4.38	8.03	18.25	40.88
NDCG@K	75.26	36.89	50.67	71.48
Precision@K	66.67	27.27	40.00	67.86
LIFT	1522.22	339.67	219.20	166.01

Panel B. Test year 2018

	pub_top_5pct	pub_top_10pct	pub_top_20pct	pub_top_30pct
Subpanel A: Human Selection Model				
Mean DV	4.07	15.45	39.84	39.84
NDCG	33.92	18.65	57.43	57.43
Precision@K	20.00	10.53	55.10	55.10
LIFT	492.00	68.14	138.32	138.32
Subpanel B: Model ABC				
Mean DV	4.07	15.45	39.84	39.84
NDCG	48.52	35.64	53.97	52.50
Precision@K	40.00	26.32	53.06	53.06
LIFT	984.00	170.36	133.19	133.19

Notes: This table compares the prediction performance of Model ABC and the Human Selection Model after excluding candidates initially placed at the top 10 U.S. business schools. The Human Selection Model uses the decile ranking of the candidate's initial placement institution as the sole predictor for research productivity in the restricted sample. Panel A reports test year 2017, and Panel B reports test year 2018. The reported results for Model ABC are based on the same prediction model as in Table 3, but all performance metrics are recalculated using the reduced test sample. Mean DV is the unconditional mean of the dependent variable in the restricted test sample. NDCG@K, Precision@K, and LIFT are defined in Appendix C.

Table 7. Prediction performance of rookie candidates' citation-weighted research productivity

Panel A. Test year 2017

Predictor	Mean DV	NDCG@K	Precision@K	LIFT
Subpanel A: DV = pub_w_top_5pct				
A	4.83	22.91	28.57	591.84
B	4.83	24.38	28.57	591.84
C	4.83	29.18	28.57	591.84
ABC	4.83	29.18	28.57	591.84
Subpanel B: DV = pub_w_top_10pct				
A	9.66	42.95	28.57	295.92
B	9.66	43.09	28.57	295.92
C	9.66	23.81	28.57	295.92
ABC	9.66	57.39	50.00	517.86
Subpanel C: DV = pub_w_top_20pct				
A	20.00	44.21	37.93	189.66
B	20.00	60.63	51.72	258.62
C	20.00	47.38	48.28	241.38
ABC	20.00	60.41	51.72	258.62
Subpanel D: DV = pub_w_top_30pct				
A	29.66	58.98	53.49	180.37
B	29.66	64.44	55.81	188.21
C	29.66	47.00	48.84	164.68
ABC	29.66	72.71	65.12	219.58

Panel B. Test year 2018

Predictor	Mean DV	NDCG@K	Precision@K	LIFT
Subpanel A: DV = pub_w_top_5pct				
A	4.58	45.39	33.33	727.78
B	4.58	41.97	33.33	727.78
C	4.58	49.35	33.33	727.78
ABC	4.58	64.48	50.00	1091.67
Subpanel B: DV = pub_w_top_10pct				
A	9.92	24.73	23.08	232.54
B	9.92	55.81	53.85	542.60
C	9.92	36.38	30.77	310.06
ABC	9.92	65.50	61.54	620.12
Subpanel C: DV = pub_w_top_20pct				
A	20.61	36.86	29.63	143.76
B	20.61	42.79	40.74	197.67
C	20.61	44.70	37.04	179.70
ABC	20.61	54.19	44.44	215.64
Subpanel D: DV = pub_w_top_30pct				
A	29.77	36.48	33.33	111.97
B	29.77	49.61	38.46	129.19
C	29.77	45.64	41.03	137.80
ABC	29.77	46.00	38.46	129.19

Notes: This table reports the out-of-sample prediction performance of models that forecast rookie candidates' citation-weighted research productivity. The dependent variables are *pub_w_top_5pct, pub_w_top_10pct, pub_w_top_20pct,* and *pub_w_top_30pct*, which are defined using candidates' citation-weighted number of top-journal publications over the five-year post-graduation window. $Num\_Pub\_Weighted_i = Num\_Pub_i * \frac{Citation_i}{\sum_{k=1}^{\#rookies\ of\ the\ same\ batch} Citation_k}$, where Num_Pub_i is the number of top journal publications for candidate *i*, and $Citation_i$ denotes the number of citations to all top-journal publications by candidate i. Panel A reports results for test year 2017, and Panel B reports results for test year 2018. Mean DV is the unconditional mean of the dependent variable in the test sample. NDCG@K, Precision@K, and LIFT are defined in Appendix C.

Table 8. Comparing the prediction performance of Model ABC and Human Selection Model for citation-weighted research productivity

Panel A. Test year 2017

	pub_w_top_5pct	pub_ w_top_10pct	pub_ w_top_20pct	pub_ w_top_30pct
	Subpanel A: Human Selection Model			
Mean DV	4.83	9.66	20.00	29.66
NDCG@K	35.81	37.10	43.79	54.10
Precision@K	28.57	35.71	37.93	44.19
LIFT	591.84	369.90	189.66	149.00
	Subpanel B: Model ABC			
Mean DV	4.83	9.66	20.00	29.66
NDCG@K	29.18	57.39	60.41	72.71
Precision@K	28.57	50.00	51.72	65.12
LIFT	591.84	517.86	258.62	219.58

Panel B. Test year 2018

	pub_ w_top_5pct	pub_ w_top_10pct	pub_ w_top_20pct	pub_ w_top_30pct
	Subpanel A: Human Selection Model			
Mean DV	4.58	9.92	20.61	29.77
NDCG@K	41.58	37.50	46.81	54.85
Precision@K	50.00	38.46	33.33	46.15
LIFT	1091.67	387.57	161.73	155.03
	Subpanel B: Model ABC			
Mean DV	4.58	9.92	20.61	29.77
NDCG@K	64.48	65.50	54.19	46.00
Precision@K	50.00	61.54	44.44	38.46
LIFT	1091.67	620.12	215.64	129.19

Notes: This table compares the out-of-sample prediction performance of Model ABC and the Human Selection Model using citation-weighted research productivity as the outcome variable. The dependent variables are *pub_w_top_5pct, pub_w_top_10pct, pub_w_top_20pct,* and *pub_w_top_30pct*. The Human Selection Model uses the decile ranking of the candidate's initial placement institution as the sole predictor. Panel A reports test year 2017, and Panel B reports test year 2018. The results for Model ABC are copied from Table 7. Mean DV is the unconditional mean of the dependent variable in the test sample. NDCG@K, Precision@K, and LIFT are defined in Appendix C.

Table 9. The 10 most important features in Model ABC and the Initial Placement Model

Panel A. Test year 2017

	Subpanel A: pub_top_5pct and job_top_5pct					
	pub_top_5 pct model	sign of pub_top_5pct model	job_top_5 pct model	sign of job_top_5pct model	Difference in ranking	Same sign
number of awards	1	+	9	+	-8	Yes
reference_high	2	-	89	+	-87	No
number of top R&R papers	3	+	168	+	-165	Yes
number of reviewers	4	+	48	+	-44	Yes
coauthor_mean	5	-	248	+	-243	No
number of coauthors	6	+	38	-	-32	No
number of membership	7	-	19	-	-12	Yes
reference_mean	8	+	11	+	-3	Yes
number of teaching experiences	9	-	26	+	-17	No
number of R&R papers	10	+	18	-	-8	No
number of published papers	13	+	7	+	6	Yes
reference_first	14	-	1	-	13	Yes
PhD_top	17	+	4	+	13	Yes
PrimaryResearchArea_financial	20	+	6	+	14	Yes
coauthor_high	58	+	5	-	53	No
number_of_coauthor_2degree	65	+	2	-	63	No
had non-academic work	88	-	8	+	80	No
number of presentations	110	-	10	+	100	No
Master_top	133	-	3	+	130	No
	Subpanel B: pub_top_10pct and job_top_10pct					
	pub_top_10pct model	sign of pub_top_10pct model	job_top_10pct model	sign of job_top_10pct model	Difference in ranking	Same sign
number of awards	1	+	8	+	-7	Yes
reference_first	2	-	1	-	1	Yes
number of coauthors	3	+	46	-	-43	No
number of membership	4	-	3	-	1	Yes
reference_high	5	-	16	+	-11	No

PhD_top	6	+	18	+	-12	Yes
number of reviewers	7	+	25	+	-18	Yes
number of teaching experiences	8	+	10	+	-2	Yes
reference_mean	9	+	5	+	4	Yes
number of top R&R papers	10	+	100	-	-90	No
number of presentations	12	-	6	+	6	No
coauthor_high	13	-	2	-	11	Yes
number of presentations at top conferences	18	-	4	+	14	No
number of papers in progress	31	+	7	+	24	Yes
PrimaryResearchArea_financial	276	-	9	+	267	No
Subpanel C: pub_top_20pct and job_top_20pct						
	pub_top_20pct model	sign of pub_top_20pct model	job_top_20pct model	sign of job_top_20pct model	Difference in ranking	Same sign
number of coauthors	1	+	12	+	-11	Yes
number of membership	2	-	5	-	-3	Yes
reference_first	3	-	14	-	-11	Yes
PhD_top	4	+	3	+	1	Yes
number of teaching experiences	5	+	4	+	1	Yes
coauthor_high	6	-	15	-	-9	Yes
coauthor_top	7	-	13	-	-6	Yes
number of reviewers	8	+	260	-	-252	No
reference_high	9	-	7	+	2	No
number of working experiences	10	-	6	+	4	No
number of awards	13	+	2	+	11	Yes
reference_mean	32	+	8	-	24	No
Bachelor_top	50	+	9	+	41	Yes
number of presentations at top conferences	76	+	10	+	66	Yes
number of presentations	85	-	1	+	84	No
Subpanel D: pub_top_30pct and job_top_30pct						
	pub_top_30pct model	sign of pub_top_30pct model	job_top_30pct model	sign of job_top_30pct model	Difference in ranking	Same sign

number of coauthors	1	+	9	+	-8	Yes
number of membership	2	-	6	-	-4	Yes
reference_first	3	-	4	-	-1	Yes
PhD_top	4	+	31	+	-27	Yes
number_of_coauthor_2degree	5	-	10	+	-5	No
number of top R&R papers	6	+	294	+	-288	Yes
Master_top	7	+	251	+	-244	Yes
coauthor_top	8	-	100	-	-92	Yes
coauthor_high	9	-	13	-	-4	Yes
number of presentations	10	+	5	+	5	Yes
number of awards	12	+	2	+	10	Yes
reference_high	14	-	1	-	13	Yes
reference_mean	15	+	3	+	12	Yes
coauthor_mean	16	+	8	+	8	Yes
number of teaching experiences	40	+	7	+	33	Yes

Panel B. Test year 2018

	Subpanel A: pub_top_5pct and job_top_5pct					
	pub_top_5pct model	sign of pub_top_5pct model	job_top_5pct model	sign of job_top_5pct model	Difference in ranking	Same sign
reference_high	1	-	1	-	0	Yes
number of awards	2	+	23	+	-21	Yes
number of reviewers	3	+	18	+	-15	Yes
number of coauthors	4	+	39	-	-35	No
reference_mean	5	+	13	+	-8	Yes
reference_first	6	-	248	-	-242	Yes
number of membership	7	-	5	-	2	Yes
number of teaching experiences	8	-	3	-	5	Yes
number of top published papers	9	+	2	+	7	Yes
number of top R&R papers	10	+	15	+	-5	Yes
number of presentations	11	+	4	+	7	Yes
number of published papers	12	+	7	+	5	Yes
number_of_coauthor_2degree	14	+	6	-	8	No
PrimaryResearchArea_financial	21	+	8	+	13	Yes

PrimaryResearchMethod_experiment	264	-	9	-	255	Yes
had non-academic work	291	+	10	+	281	Yes
			Subpanel B: pub_top_10pct and job_top_10pct			
	pub_top_10pct model	sign of pub_top_10pct model	job_top_10pct model	sign of job_top_10pct model	Difference in ranking	Same sign
reference_first	1	-	18	-	-17	Yes
number of awards	2	+	3	+	-1	Yes
number of coauthors	3	+	10	+	-7	Yes
number of membership	4	-	4	-	0	Yes
number of reviewers	5	+	16	+	-11	Yes
coauthor_high	6	-	11	-	-5	Yes
number of top published papers	7	+	8	+	-1	Yes
reference_high	8	-	1	-	7	Yes
number of top R&R papers	9	+	13	+	-4	Yes
reference_mean	10	+	7	+	3	Yes
PhD_top	12	+	5	+	7	Yes
number_of_coauthor_2degree	13	+	9	-	4	No
number of working experiences	15	-	6	+	9	No
number of presentations	18	+	2	+	16	Yes
			Subpanel C: pub_top_20pct and job_top_20pct			
	pub_top_20pct model	sign of pub_top_20pct model	job_top_20pct model	sign of job_top_20pct model	Difference in ranking	Same sign
number of coauthors	1	+	16	+	-15	Yes
number of presentations	2	+	1	+	1	Yes
PhD_top	3	+	3	+	0	Yes
reference_first	4	-	4	-	0	Yes
coauthor_high	5	-	13	-	-8	Yes
number of membership	6	-	6	-	0	Yes
coauthor_mean	7	+	7	+	0	Yes
number of teaching experiences	8	+	54	+	-46	Yes
number of reviewers	9	+	19	+	-10	Yes
number of awards	10	+	2	+	8	Yes
number of top published papers	12	+	9	+	3	Yes
number of presentations at top conferences	15	-	10	+	5	No

reference_mean	18	-	8	-	10	Yes
number of working experiences	159	-	5	+	154	No
			Subpanel D: pub_top_30pct and job_top_30pct			
	pub_top_30pct model	sign of pub_top_30pct model	job_top_30pct model	sign of job_top_30pct model	Difference in ranking	Same sign
number of coauthors	1	+	36	+	-35	Yes
reference_first	2	-	22	+	-20	No
number of membership	3	-	3	-	0	Yes
coauthor_high	4	-	55	-	-51	Yes
PhD_top	5	+	1	+	4	Yes
coauthor_mean	6	+	222	-	-216	No
number of presentations	7	+	11	+	-4	Yes
number of awards	8	+	4	+	4	Yes
Master_top	9	+	182	-	-173	No
number of top R&R papers	10	+	18	+	-8	Yes
number of reviewers	12	+	10	+	2	Yes
number of top published papers	16	+	5	+	11	Yes
number_of_coauthor_2degree	18	-	6	-	12	Yes
gender	23	+	7	+	16	Yes
reference_high	26	-	2	-	24	Yes
number of published papers	29	+	9	+	20	Yes
has Bachelor honor	271	-	8	+	263	No

Notes: This table reports the 10 most important input variables, ranked by the magnitude of SHAP values, for Model ABC and the Initial Placement Model in the test-year samples. Panel A reports test year 2017, and Panel B reports test year 2018. For each dependent variable, rank 1 denotes the most important feature. The sign reported for each feature is determined in two steps. First, we regress the SHAP value of the feature on the corresponding feature value in the test sample. Then, we assign the sign of the estimated coefficient from that regression. If the absolute value of the estimated coefficient is smaller than 1% of the unconditional mean of the predicted dependent variable, we treat the feature as having no meaningful economic effect and assign a value of zero.

Table 10. Comparing *SHARE_UNADJ* and *SHARE_ADJ* of ML Selected Group and Human Hired Group

Test year	*SHARE_UNADJ*	*SHARE_ADJ*
	Panel A: Human Hired Group	
2017	17.19%	18.55%
2018	22.77%	24.96%
	Panel B: ML Selected Group	
2017	31.25%	30.58%
2018	25.74%	30.05%

Notes: This table compares the publication shares of the Human Hired Group and the ML Selected Group. The Human Hired Group comprises candidates hired by the top 10 U.S. business schools in each test year. The ML Selected Group consists of the same number of candidates, selected according to the highest predicted probabilities from Model ABC using the pub_top_5pct specification. *SHARE_UNADJ* is the share of top-journal publications produced by the selected group relative to all candidates in the same cohort during the five-year post-graduation window. *SHARE_ADJ* is the corresponding share based on citation-weighted publications.

Table 11. OLS regression results of *gap_Kpct_prob* on rookie candidates' characteristics (test years 2017 and 2018 combined)

Panel A. Descriptive statistics of *gap_Kpct_prob*

	K=5	K=10	K=20	K=30
count	276	276	276	276
mean	0.056	0.011	-0.017	-0.009
std	0.153	0.086	0.078	0.088
min	-0.496	-0.255	-0.271	-0.269
25%	-0.039	-0.049	-0.062	-0.073
50%	0.044	0.026	-0.010	-0.014
75%	0.133	0.076	0.035	0.054
max	0.557	0.201	0.186	0.187

Panel B. OLS regression results and *t* stats in parenthesis

	K=5	K=10	K=20	K=30
gender	-0.0359*	-0.0090	0.0097	-0.0040
	(-1.92)	(-0.86)	(1.09)	(-0.39)
second_language_asia	-0.0167	-0.0036	0.0051	0.0014
	(-0.48)	(-0.19)	(0.31)	(0.08)
second_language_euro	-0.0260	-0.0107	-0.0291**	-0.0551***
	(-0.94)	(-0.69)	(-2.20)	(-3.66)
PhD_top10	-0.0373	-0.0281*	-0.0250*	-0.0261
	(-1.22)	(-1.64)	(-1.71)	(-1.58)
reference_top	0.0015	0.0174*	0.0492***	0.0403***
	(0.08)	(1.66)	(5.56)	(4.00)

Notes: This table reports descriptive statistics and OLS regression results for *gap_Kpct_prob* using the pooled test-year sample from 2017 and 2018. *gap_Kpct_prob* is defined as the predicted probability of *pub_top_Kpct* from Model ABC minus the predicted probability of *job_top_Kpct* from the Initial Placement Model. The Initial Placement Model uses the same predictor sets A, B, and C as Model ABC but predicts the likelihood that the candidate's initial placement institution ranks in the top K percent. Panel A reports descriptive statistics. Panel B reports OLS coefficient estimates, with t-statistics in parentheses. *gender* equals one if the applicant is male and zero otherwise. *second_language_asia* equals one if the applicant reports proficiency in an Asian language other than English, and zero otherwise. *second_language_euro* equals one if the applicant reports proficiency in a European language other than English, and zero otherwise. *PhD_top10* equals one if the applicant received the PhD from a top-10 business school, and zero otherwise. *reference_top* equals one if the applicant's first reference is ranked in the top 10 percent of the BYU individual ranking in that year, and zero otherwise.

Appendix A. Variable definitions

variable name	data source	definition
	Panel A: dependent variables	
pub_top_5pct	hand-collect	The dummy indicating whether the candidate's number of publications on the top 11 journals (listed in the data source section) during the 5 years starting from the batch year (e.g., 2018.1~2022.12 for the 2017/2018 batch) is among the top 5% of his/her peers who apply to the Miami Camp in the same year
pub_top_10pct	hand-collect	The dummy indicating whether the candidate's number of publications on the top 11 journals (listed in the data source section) during the 5 years starting from the batch year (e.g., 2018.1~2022.12 for the 2017/2018 batch) is among the top 10% of his/her peers who apply to the Miami Camp in the same year
pub_top_20pct	hand-collect	The dummy indicating whether the candidate's number of publications on the top 11 journals (listed in the data source section) during the 5 years starting from the batch year (e.g., 2018.1~2022.12 for the 2017/2018 batch) is among the top 20% of his/her peers who apply to the Miami Camp in the same year
pub_top_30pct	hand-collect	The dummy indicating whether the candidate's number of publications on the top 11 journals (listed in the data source section) during the 5 years starting from the batch year (e.g., 2018.1~2022.12 for the 2017/2018 batch) is among the top 30% of his/her peers who apply to the Miami Camp in the same year
job_top_5pct	hand-collect; US News; QS	The dummy indicates whether the candidate's first placement university is among the top 5% of his/her peers who apply to the Miami Camp in the same year. The university rank is the combined historical rank from U.S. News and QS World University Rankings.
job_top_10pct	hand-collect; US News; QS	The dummy indicates whether the candidate's first placement university is among the top 10% of his/her peers who apply to the Miami Camp in the same year. The university rank is the combined historical rank from U.S. News and QS World University Rankings.
job_top_20pct	hand-collect; US News; QS	The dummy indicates whether the candidate's first placement university is among the top 20% of his/her peers who apply to the Miami Camp in the same year. The university rank is the combined historical rank from U.S. News and QS World University Rankings.
job_top_30pct	hand-collect; US News; QS	The dummy indicates whether the candidate's first placement university ranks among the top 30% of his/her peers who apply to the Miami Camp in the same year. The university rank is the combined historical rank from U.S. News and QS World University Rankings.

Panel B: variables extracted directly from the CV without background knowledge (Predictors A)		
gender	CV	1 for male, 0 for female
has Bachelor honor	CV	A dummy indicating whether the candidate listed any honors obtained during the Bachelor's stage in the CV
has Master honor	CV	A dummy indicating whether the candidate listed any honors obtained during the Master's stage in the CV
has PhD honor	CV	A dummy indicating whether the candidate listed any honors obtained during the Ph.D. stage in the CV
number of published papers	CV	The total number of published papers listed in the CV
number of R&R papers	CV	The total number of papers that got an R&R listed in the CV
number of papers in progress	CV	The total number of papers in progress listed in the CV
has_coauthor	CV	The dummy indicating whether the candidate mentioned any coauthor in the CV
has_reference	CV	The dummy indicates whether the candidate mentioned any reference in the CV
number of coauthors	CV	The number of coauthors mentioned in the CV
number of presentations	CV	The total number of conference or workshop presentations listed in the candidate's CV
number of teaching experiences	CV	The total number of teaching experiences listed in the CV
number of awards	CV	The total number of academic awards listed in the CV
number of reviewers	CV	The number of journals or associations where the candidate serves as a reviewer listed in the CV
number of membership	CV	The number of memberships listed in the CV
number of working experiences	CV	The number of working experiences listed in the CV
had academic work	CV	The dummy indicates whether the candidate had academic employment experience mentioned in the CV
had non-academic work	CV	The dummy indicates whether the candidate had non-academic working experience mentioned in the CV
provide abstract	CV	The dummy indicates whether the candidate presents abstracts of any papers in the CV
PrimaryResearchArea_financial	CV	A dummy indicating that the candidate’s primary research area is financial accounting
PrimaryResearchArea_auditing	CV	A dummy indicating that the candidate’s primary research area is auditing
PrimaryResearchArea_tax	CV	A dummy indicating that the candidate’s primary research area is tax
PrimaryResearchArea_managerial	CV	A dummy indicating that the candidate’s primary research area is managerial accounting
PrimaryResearchMethod_archival	CV	A dummy indicating that the candidate’s primary research method is archival empirical methods

PrimaryResearchMethod_experimental	CV	A dummy indicating that the candidate's primary research method is experimental methods
PrimaryResearchMethod_analytical	CV	A dummy indicating that the candidate's primary research method is analytical methods
multi_language	CV	A dummy indicating that the candidate speaks more than one language

Panel C: variables extracted from CV combined with background knowledge (Predictors B)		
bachelor_top	CV; US News	A dummy indicating whether the candidate earned a bachelor's degree from one of the top 50 business schools in the US. The university name is from the "Bachelor's degree" item output by GPT. The rank is the historical university rank from US News.
master_top	CV; US News	A dummy indicating whether the candidate earned a master's degree from one of the top 50 business schools in the US. The university name is from the "Master's degree" item output by GPT. The rank is the historical university rank from US News.
phd_top	CV; US News	A dummy indicating whether the candidate earned the PhD degree from one of the top 50 business schools in the US. The university name is from the "PhD degree" item output by GPT. The rank is the historical university rank from US News.
visit_top	CV; US News	A dummy indicating whether the candidate has ever visited the top 50 business schools in the US. The university names are from the "visiting experience" item output by GPT. The rank is the historical university rank from US News.
number of top published papers	CV; field knowledge	The total number of papers that are listed in the CV and published in the 11 top journals (listed in the data source section). The journal names are from the "published journal list" by GPT.
number of top R&R papers	CV; field knowledge	The total number of papers that are listed in the CV and received an R&R in the 11 top journals (listed in the data source section). The journal names are from the "R&R journal list" by GPT.
coauthor_mean	CV; BYU	The average rank of all the coauthors mentioned in the CV. The rank is from the historical individual rank from the BYU website in the year prior to the candidate's batch year. The coauthor names are from the "coauthor list" item by GPT. If no coauthor is mentioned, fill 0.[11]
coauthor_high	CV; BYU	The highest rank of all the coauthors mentioned in the CV. The rank is from the historical individual rank from the BYU website in the year prior to the candidate's batch year. The coauthor names are from the "coauthor list" item by GPT. If no coauthor is mentioned, fill 0.

[11] We define a variable *has_coauthor* to identify the missing cases.

number_of_coauthor_2nd_degree [12]	CV; BYU	The total number of coauthors of all the coauthors mentioned in the CV. The coauthors of each coauthor are from the BYU website page for each individual accounting scholar. If the coauthor does not have a page on the BYU website, we assume the coauthor is not well known and do not count them. The coauthor names are from the "coauthor list" item by GPT.
coauthor_top	CV; BYU	A dummy of whether any one of the candidate's coauthors mentioned in the CV is a top scholar (top scholar is defined as the scholar ranked in the top 1% on the BYU website in the application year). The coauthor names are from the "coauthor list" item by GPT.
number of top conference presentation	CV; field knowledge	The total number of presentations at top accounting conferences, including the American Accounting Association (AAA) annual conference, the Journal of Accounting and Economics (JAE) conference, the Contemporary Accounting Research (CAR) conference, the Journal of Accounting Research (JAR) conference, Review of Accounting Studies (RAS) conference, Financial Accounting and Reporting Midyear meeting, Managerial Accounting Midyear Meeting, Journal of the American Taxation Association (JATA) conference.
reference_first	CV; BYU	The rank of the first name in the reference list in the year prior to the candidate's batch year. The rank is the historical individual rank from the BYU website. The reference names are from the "references" item by GPT. If no reference name is mentioned, fill 0.[13]
reference_mean	CV; BYU	The average rank of the reference names in the year prior to the candidate's batch year. The rank is the historical individual rank from the BYU website. The reference names are from the "references" item by GPT. If no reference name is mentioned, fill 0.
reference_high	CV; BYU	The highest rank of the reference names in the year prior to the candidate's batch year. The rank is the historical individual rank from the BYU website. The reference names are from the

[12] n*umber_of_coauthor*, *number_of_coauthor_2nd_degree*, and coauthor ranks (i.e., *coauthor_top, coauthor_high, coauthor_mean*) are all network related variables. Ductor et al. (2014) also include measures such as "closeness centrality" and "betweenness centrality". We didn't include these measures for two reasons: 1. Their work focuses on all economic scholars, but we only focus on rookies. Rookies usually do not form very complex coauthorship networks, so these measures are highly correlated with existing variables. They may not add much incremental information. 2. Their work constructs a network from EconLit. Our coauthorship data come from individual CVs and the BYU website, which includes only prestigious scholars. So our network is incomplete, making these measures inaccurate and less relevant.

[13] We define a variable *has_reference* to identify the missing cases.

		"references" item by GPT. If no reference name is mentioned, fill 0.
second_language_Asia	CV; knowledge	A dummy indicating that the candidate speaks a second language besides English and that the language is from an Asian country.
second_language_Euro	CV; knowledge	A dummy indicating that the candidate speaks a second language besides English and that the language is from a European country.

Panel D: embeddings from the dissertation by Text-Embedding-3-Large (Predictors C)		
1_dt ~ 256_dt	dissertation	256 variables obtained by applying Text-Embedding-3-Large to the dissertations' title, abstract, and introduction section, and only keeping the first 256 numbers of the embeddings.

Panel E: initial placement rank		
Placerank	US News; QS University Rankings	This is the rank of the candidate's first placement institution. Each year, we first identify the top 80 schools based on the combined rankings from U.S. News and QS. Then we rank the placement schools of rookies. In each year, for the samples that get jobs at the top 80 schools, we group them into deciles by rank and define the Place_rank variable as 1-10, where 1 denotes the top schools. For the samples that don't have a job on the top 80 schools, their Place_rank variable is set to 11.

Appendix B. The IPW procedure

We implement the inverse probability weighting (IPW) procedure following Swaminathan and Joachims (2015) and Schnabel et al. (2016). Because the method requires a dichotomous treatment indicator, we classify initial placement institutions as either research-oriented or non-research-oriented. We assume that institutions within each category are relatively homogeneous in research intensity.

We determine research orientation using region-specific classification systems to ensure comparability across countries. For institutions located in the United States, we rely on the Carnegie Classification of Institutions of Higher Education. Specifically, we designate an institution as research-oriented if it qualifies as a Research 1 (R1) university, defined as spending at least USD 50 million on research and development (R&D) and awarding at least 70 research doctorates annually.

For institutions located in Europe, we use the European Tertiary Education Register (ETER), which provides standardized institution-level information on doctoral education and research expenditures. To maintain consistency with the U.S. Carnegie criteria, we classify a European institution as research-oriented if it awards more than 70 doctoral degrees per year and reports annual R&D expenditures exceeding USD 50 million.

For institutions outside the United States and Europe, we identify research-oriented universities using widely recognized national classifications of research-intensive institutions. In mainland China, we classify "985" universities and "211/Double First Class" universities as research-oriented. In Taiwan, we classify National Taiwan University as research-oriented. In South Korea, we include Seoul National University, KAIST, POSTECH, SKKU, Yonsei University, and Korea University. In

Australia, we include the Group of Eight (Go8) universities. In Singapore, we include NUS, NTU, and SMU. In Hong Kong, we classify the five leading research universities—HKU, CUHK, HKUST, PolyU, and CityU—as research-oriented. In Canada, we include the U15 Research Universities. All remaining institutions in our sample are classified as non-research-oriented.

We next estimate a logit model of a candidate's initial placement type. The dependent variable, D_i, is an indicator equal to one if the candidate's initial placement institution is classified as research-oriented and zero otherwise. We model $D_i$as a function of the predictor set used in the machine learning model. For instance, if the machine learning model uses predictor set ABC, we also use ABC in the logit model.

The model is estimated separately for each test year using the corresponding training sample. The logit specification is as follows:

$$\Pr(D_i = 1 | X_i) = e(X_i; \gamma)$$

To ensure that the IPW procedure is well behaved, we first examine the empirical distribution of the estimated propensity scores, $\hat{e}(X_i)$, and trim observations with $\hat{e}(X_i)$ outside the interval ([0.05, 0.95]) (Crump, Hotz, Imbens, and Mitnik 2009; Austin and Stuart 2015). Candidates with extreme propensity scores are effectively deterministically assigned to one institutional type, conditional on X_i, and their inclusion would produce excessively large weights and increase the estimator's variance.

As a diagnostic, we evaluate common support and covariate balance by comparing standardized mean differences in the primary predictors X_i between research- and non-research-oriented institutions, both before and after applying the IPW weights. We consider the IPW adjustment successful if the absolute standardized differences decline substantially after weighting, indicating improved balance across treatment groups. We find that IPW adjustment improves overall covariate balance across treatment groups (untabulated).

Next, we use the estimated propensity scores to construct stabilized inverse probability weights that generate a pseudo-population in which institutional type is independent of candidate characteristics, conditional on X_i. Let $\hat{p} = \frac{1}{N}\sum_i D_i$ denote the empirical fraction of candidates placed at research-oriented institutions in the training sample for a given test year. For each candidate i, we define the stabilized weight following Hernán and Robins (2020) as follows:

$$w_i = \frac{\hat{p}}{\hat{e}(X_i)} D_i + \frac{1 - \hat{p}}{1 - \hat{e}(X_i)} (1 - D_i)$$

Intuitively, candidates whose realized institutional type is unlikely given their observed characteristics—for example, high-potential candidates placed at non-research-oriented institutions or lower-potential candidates placed at research-oriented institutions—receive larger weights. These observations are particularly informative about how research productivity varies with X_i when the assignment to institutional type is less strongly aligned with observable characteristics. To further reduce the influence of extreme weights, we truncate w_i at the 5th and 95th percentiles of its empirical distribution. Finally, we re-estimate our research productivity prediction models using the stabilized IPW weights.

Appendix C. Prediction performance metrics

metrics	definition
Mean DV	The unconditional mean of the dependent variable
NDCG@K	Normalized Discounted Cumulated Gain at the position K. K is equal to the unconditional mean of the dependent variable in the test set
Precision@K	The percentage of candidates among the top K ranked by predicted probabilities in a given test year whose actual research productivity falls within the top K of candidates in the test year
LIFT	Precision@K as a percentage of Mean DV, representing the accuracy of a prediction model relative to a random classification benchmark

Online appendix

Table OA1. Prompting Using GPT-4

As a recruitment member of the accounting department of a university. Your task is to process the CV texts of candidates, extract the information I want to structural output. Here is the list of the information I want from their CVs.

1. Name
2. Education experience
3. Employment experience
4. Research interest
5. Publication
6. Working paper
7. Conference presentation
8. Teaching experience
9. Awards
10. Professional service and membership
11. Reference
12. Language

Present your findings as a single JSON object, conforming to the following structure:

```
{
  "name": "name of the candidate",
  "Bachelor degree": "the name of the university where the candidate got the bachelor degree (Do not include the country. Do not include the school name or department name, just the name of the university.)",
  "has Bachelor honor": 1 if the candidate has an honor in bachelor stage, 0 otherwise,
  "Master degree":  "the name of the university where the candidate got the master degree (do not include the country. Do not include the school name or department name, just the name of the university.)",
  "has Master honor": 1 if the candidate has an honor in Master stage, 0 otherwise,
  "PhD degree": "the name of the university where the candidate will get the PhD degree (do not include the country. Do not include the school name or department name, just the name of the university.)",
  "has PhD honor": 1 if the candidate has an honor (e.g., Summa Cum Laude) in PhD stage, 0 otherwise,
  "visiting experience": the list of universities (note: not school names) the candidate visited as a visiting PhD or visiting scholar(do not include the country), in the form of a list ["university 1","university 2",...],
  “research interest”: the text describing the research interests of the candidate,
  "papers": the list of paper including the following attributes: title,coauthor (if no coauthor, say "solo"), journal (if not under review, leave this blank), status (including under review, revise and resubmit(R&R), published),
  "number of published papers": the number of published papers, 0 if no publication,
  "published journal list": the list of the journals if the candidate has published papers like ["journal 1", "journal 2"], blank list if no publication,
  "number of R&R papers": the number of papers that get revise and resubmit (R&R) by journals (note that under review is not R&R),
  "R&R journal list": the list of the journals gave an R&R to the candidate like ["journal 1", "journal 2"],
  "number of papers in progress": the number of papers that are not published or get an R&R,
  "coauthor list": the list of all the coauthors of all the papers of the candidate like ["coauthor 1","coauther 2",...],
  “number of coauthors”: the number of coauthors of all the papers mentioned in this CV,
  "conference presentation": the list of conferences the candidate participated (including academic conferences and workshops),
  "number of presentations": the total number of presentations, including academic conferences, workshops, etc.
  "number of presentations at top conferences": the total number of presentations at top conferences, including American Accounting Association (AAA) annual conference, Journal of Accounting and Economics (JAE) conference, Contemporary Accounting Research (CAR) conference, Journal of Accounting Research (JAR) conference, Review of Accounting Studies (RAS) conference, Financial Accounting and Reporting Midyear meeting, Managerial Accounting Midyear Meeting, Journal of the American Taxation Association (JATA) conference.
  "Teaching experiences": the list of teaching experiences in the form of ["experience 1","experience 2",......],
  "number of teaching experiences": the number of teaching experiences,
  "academic awards": the list of academic awards,
  “number of awards”: the number of academic awards,
  "reviewer": the list of journals or associations where the candidate serves as ad hoc reviewer,
  “number of reviewers”: the number of journals or associations where the candidate serves as ad hoc reviewer,
  "membership": the associations that the candidate join as a member,
  “number of membership”: the number of associations that the candidate join as a member,
  "number of working experiences": the number of working experiences,
  "had academic work": 1 if the candidate has worked in academic institutions, 0 otherwise,
  "had non-academic work": 1 if the candidate has worked outside academic institutions, 0 otherwise,
  "references": the list of names of the references like ["name 1","name 2"] (remove the titles like professor, prof, Dr, etc.),
  "provide abstract": 1 if the candidate presents the abstracts of the papers in the CV file, 0 otherwise,
  "abstract list": the list of the abstracts in the form of ["abstract 1", "abstract 2",......]
  “language list”: the list of languages that the candidate speak in the form of [“language 1”,”language 2”,…], blank if not mentioned.
}
```

If any of the above information is not mentioned in the CV profile, just fill the item with a blank. Your output should be a single valid JSON object and should not contain any additional or explanatory text. Here is the text of the candidate CV:

Assessing the accuracy of GPT's feature extraction from resumes

To evaluate the accuracy of GPT in converting resumes into structured variables, we randomly selected 20 samples from all CV files and tasked a research assistant (RA) with manually extracting attributes from these CVs using the same prompting format shown in Table OA1. We then compared the numeric predictors derived from GPT to those extracted by the RA. Table OA2 summarizes the discrepancies between GPT's outputs and the RA's results, including the number of mismatched samples and the mean absolute error (MAE) for each variable.

Of the 17 variables analyzed, GPT achieved perfect extraction accuracy for six and made fewer than ten errors for an additional nine. Only two variables—*the number of papers in progress* and *the number of presentations*—had error rates exceeding 50%. These inaccuracies likely reflect inconsistencies and lack of transparency in how applicants disclose this information, which complicates accurate extraction. Despite these discrepancies, the mean absolute errors are generally small and statistically insignificant for most predictors, indicating that GPT's extraction performance is acceptable for the majority of variables.

Table OA2. Accuracy of 20 randomly selected samples: comparing GPT with human RAs

	#(GPT≠RA)	Mean Absolute Error
has Bachelor honor	0	0
has Master honor	0	0
has PhD honor	0	0
had non-academic work	0	0
provide abstract	0	0
multi_language	0	0
number of R&R papers	2	0.10
number of published papers	3	0.35
number of membership	3	0.25
number of reviewers	4	0.50
number of awards	6	0.50
number of coauthors	7	0.65
number of working experiences	7	0.55
number of teaching experiences	8	1.20
had academic work	9	0.45
number of papers in progress	12	1.35
number of presentations	12	3.10

Notes: This table compares the structured variables extracted by GPT with those manually extracted by a research assistant from 20 randomly selected CVs. The first column reports the number of cases in which the GPT extraction differs from the research assistant's extraction. The second column reports the mean absolute error between the two extraction methods. See Appendix A for variable definitions.

Table OA3. Prediction performance of the Initial Placement Model

Panel A: test year 2017

Predictor	Mean DV	NDCG@K	Precision@K	LIFT
Subpanel A: DV = job_top_5pct				
A	4.83	11.84	14.29	295.92
B	4.83	0.00	0.00	0.00
C	4.83	0.00	0.00	0.00
ABC	4.83	11.84	14.29	295.92
Subpanel B: DV = job_top_10pct				
A	7.59	13.15	18.18	239.67
B	7.59	10.37	9.09	119.83
C	7.59	22.31	27.27	359.50
ABC	7.59	8.93	9.09	119.83
Subpanel C: DV = job_top_20pct				
A	17.24	23.67	24.00	139.20
B	17.24	31.59	32.00	185.60
C	17.24	55.65	56.00	324.80
ABC	17.24	38.06	32.00	185.60
Subpanel D: DV = job_top_30pct				
A	28.97	33.34	35.71	123.30
B	28.97	29.29	26.19	90.42
C	28.97	48.38	38.10	131.52
ABC	28.97	40.43	35.71	123.30

Panel B: test year 2018

Predictor	Mean DV	NDCG@K	Precision@K	LIFT
		Subpanel A: DV = job_top_5pct		
A	3.82	0.00	0.00	0.00
B	3.82	36.01	40.00	1048.00
C	3.82	0.00	0.00	0.00
ABC	3.82	69.92	60.00	1572.00
		Subpanel B: DV = job_top_10pct		
A	8.40	20.74	9.09	108.26
B	8.40	36.95	36.36	433.06
C	8.40	25.32	27.27	324.79
ABC	8.40	52.98	45.45	541.32
		Subpanel C: DV = job_top_20pct		
A	19.85	26.60	23.08	116.27
B	19.85	35.23	30.77	155.03
C	19.85	51.18	46.15	232.54
ABC	19.85	41.26	34.62	174.41
		Subpanel D: DV = job_top_30pct		
A	29.77	54.03	58.97	198.09
B	29.77	47.79	46.15	155.03
C	29.77	56.41	56.41	189.48
ABC	29.77	68.97	64.10	215.32

Notes: These tables report the out-of-sample prediction performance of the Initial Placement Model, which predicts whether a candidate's initial placement institution falls within the top K percent of the cohort in the same year. Panel A reports results for test year 2017, and Panel B reports results for test year 2018. Predictor sets A, B, and C and the ensemble model ABC are defined as in Table 3. Mean DV is the unconditional mean of the dependent variable in the test sample. NDCG@K, Precision@K, and LIFT are defined in Appendix C.